

Causal Coding and the General Causal-Continual-Learning Theorem

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December 1, 2025

Abstract

We present a general framework in which *continual learning* robustness is a structural consequence of learning and using correct (or approximately correct) causal structure. The key idea is that when an AI system factors its world-model into causally meaningful modules, and allocates its computational budget so as to keep those modules as independent as the data allow, then the gradient flows associated with different *contexts* almost commute. In that regime, training under a stream of changing contexts is provably close to training on a joint mixture, and catastrophic forgetting is quantitatively controlled, with explicit error terms that depend on the amount of inter-context overlap.

We formalize this as a *General Causal-Continual-Learning Theorem* in an abstract operator setting, and then give two concrete specializations: (i) a continuous-variable version on smooth parameter manifolds using gradient flows and Lie brackets; and (ii) a discrete version for general state/update systems. Both are instances of a more general category-theoretic formulation in which context-conditioned updates are endomorphisms on a causally factorized object, and “small commutators” are measured in an enriching quantale of loss and computational cost.

Finally, we specialize the theorem to (a) causal-coding versions of predictive-coding neural networks, where interventional Jacobians and clarity regularizers suppress cross-module Hessian blocks, and (b) predictive coding Transformers, where causal coding acts at the level of attention heads and value channels. In both cases we obtain explicit qualitative and quantitative conditions under which causal coding makes catastrophic forgetting provably hard to achieve, in a *probably approximately correct* sense that degrades smoothly as causal modularity assumptions are relaxed.

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1 Introduction

The aim of this paper is to give a simple, clear and practical explanation for why

- **causal coding**, considered as a general family of learning mechanisms for guiding learning-networks toward internal structure reflecting causal relations in the world (spanning e.g. both continuous and discrete systems, and neural and non-neural systems)

can be generally effective for

- **continual learning** (avoiding pathologies like “catastrophic forgetting” that one sees in typical backpropagation-based neural nets)

The high-level story is simple. Any reasonably general learner sees many *contexts* over its lifetime: different input regimes, goals, interaction patterns, or parts of the world. If adapting to a new context always re-writes the same small set of entangled parameters, then the system must forget: new updates tear up old solutions. If, instead, the system carves its internal state into reasonably independent causal modules, so that each context mostly tunes its own subset of modules while leaving the rest alone, then adapting to new contexts does not destroy competence in older ones.

The central claim of this paper is that:

Causal modeling, combined with an explicit drive to keep latent factors as independent as the data allow, is exactly what you need to make this modularity happen—and to make continual learning work.

More concretely, the argument has four conceptual steps:

1. **Contexts correspond to different ways of querying the same causal world.** If there really is a shared causal structure underlying all the contexts an agent encounters, then there is a distinguished representation in which that structure becomes a relatively sparse graph over latent variables, with many weak or absent edges. Different contexts then depend on different subgraphs.
2. **Independence vs. dependency is not optional: it is an optimization tradeoff.** A system can either (1) explicitly track dependencies (couplings, interactions, attention links), or (2) reparametrize its state to make things more independent (whitening, disentangling, abstraction). Both routes cost compute; the “tradeoff theorem” says you cannot hit a fixed accuracy target without investing in at least one of them. If you optimize this tradeoff, you end up with representations that are as factorized as the data permit.

3. **When the representation is factorized, context updates almost commute.** Each context induces a vector field in parameter space (its gradient field under the context-specific loss). If different contexts touch almost-disjoint modules, their vector fields almost commute in the Lie bracket sense. When flows commute, doing context A then B is almost the same as doing B then A, and almost the same as doing a single step on the combined loss over both contexts. When contexts overlap, commutators are controlled by the size and strength of the overlaps, as made explicit in the abstract categorical formulation and its “confusing pairs” bounds.
4. **When context flows almost commute, forgetting is bounded.** General results on compositions of flows show that the deviation between sequential training under contexts $1, \dots, T$ and a joint multi-context update is controlled by the pairwise commutators. If those commutators are small, the performance drop in earlier contexts after adapting to new ones is provably small, and the error grows at most linearly with the number of genuinely interacting context pairs.

The role of *causal coding* is to operationalize steps (1) and (2) in predictive-coding style networks. Causal coding uses interventional influence (do-derivatives) to decide which connections really carry causal signal, gates updates through those connections, and prunes redundant routes that mainly serve to diffuse influence. Together with the general independence/coupling tradeoff, this pushes the network toward a causally-meaningful, weakly entangled latent factorization. Once you have that, the commutator and continual-learning results kick in.

Very roughly, the philosophical moral is:

To do continual learning well, an AI should treat new contexts as local updates to a causal world-model that it already factorized as much as possible. Causal coding is one way—and perhaps a particularly clean way—to enforce this factorization and make the underlying gradient flows nearly commuting.

What this paper does. On the technical side we:

- Formalize an abstract setting in which a family of contexts corresponds to a family of update operators on a state/configuration space equipped with a causal factorization.
- State and prove a *General Causal-Continual-Learning Theorem* that says: if context-conditioned updates respect this causal factorization in a precise sense (only touching the modules that causally matter to them, plus small residual couplings), then their update operators almost commute, and sequential training almost matches joint multi-context training, up to error terms that depend explicitly on a sparse graph of “confusing” context pairs.
- Give two concrete specializations:
 1. a continuous-variable version on a smooth parameter manifold, with context updates given by gradient flows and “almost commute” measured by Lie brackets of vector fields;
 2. a discrete-variable version where the state space is arbitrary and context-conditioned updates are arbitrary endomorphisms, with “almost commute” measured in a metric or quantale-enriched sense.
- Show how the continuous version applies to causal-coding predictive-coding networks, where do-influence matrices and clarity regularizers suppress cross-module Hessian blocks and thus shrink commutators.

- Sketch how the same structure appears in predictive-coding Transformers, where causal coding acts across attention heads and feed-forward channels.

Paper structure. Section 2 reviews the independence/coupling tradeoff, predictive and causal coding, PC Transformers, and the commutativity-based view of continual learning. Section 3 states and proves the general Causal-Continual-Learning Theorem and gives a nontechnical interpretation. Sections 4 and 5 give continuous and discrete specializations. Sections 6 and 7 apply the theorem to causal-coding neural networks and then to PC Transformers specifically, followed by Section 8 which considers application to incremental compression (as an example of a purely discrete, non-neural application).

Appendix A gives a general categorical formulation of the Causal-Continual-Learning Theorem, and Appendix B gives some hypothetical in-depth examples of the application to transformers. Appendix C clarifies some aspects of the application to incremental compression—noting that in a best-case regime where a modular optimum exists, the use of causal coding can accelerate convergence toward this optimum without changing it; and in more typical regimes it biases learning toward different, more modular optima that may be preferable for continual learning even if they differ from the plain incremental-compression optimum.

2 Background

2.1 The independence-coupling tradeoff and weakness

Any system that processes multiple information channels faces an irreducible choice:

- *Route A: independence-enforcing transforms.* Change representation to make channels more independent (whitening, decorrelation, factorization, abstraction, masking). This reduces the amount of explicit dependency you need to track later but costs computation now.
- *Route B: explicit dependency modeling.* Keep the representation as-is, but introduce explicit couplings (factors, attention, joins, multi-ary operations) that track dependencies. This can explode in cost as higher-order interactions are added.

The “tradeoff theorem” (in its various analytic and categorical forms) says that, subject to resource constraints, any method that hits a given information-efficiency target must pay at least one of these costs. More refined statements characterize Pareto-optimal operating points as those where the marginal information gain per unit cost is equalized between Route A and Route B. Practically, this implies that a well-designed system will:

1. perform just enough independence-inducing transforms to keep the remaining dependency mass manageable; and
2. introduce explicit couplings only where the residual dependencies really matter for predictive accuracy.

This is the formal backbone behind the intuitive statement that “good architectures are as factorized as the data allow.”

2.2 Predictive coding and causal coding

Predictive coding (PC) treats a network as implementing inference in a hierarchical generative model. Each layer predicts the activity of the layer below; prediction errors are driven to zero by iterative updates of latent variables; weights are updated by local Hebbian-style rules driven by prediction errors. In the standard formulation, the key quantities are

- a free energy $F(x, z; \theta)$, often quadratic in residuals;
- inference dynamics $\dot{z} = -\partial F / \partial z$ relaxing to a fixed point;
- learning rules $\Delta \theta \propto \partial F / \partial \theta$ using local error and presynaptic activity.

Causal coding (CC) modifies this picture in two crucial ways:

1. *Causal gates.* Instead of weighting updates by mere correlation, CC estimates *do-influence* matrices C , which quantify how much a child’s equilibrium would change under a small Pearlian intervention on a parent unit. Learning updates are then gated by $|C_{ji}|^p$ (with $p \in [1, 2]$), focusing learning on connections that actually carry causal signal.
2. *Clarity / pruning.* CC penalizes redundant or diffusive routes, for example via diffusion-based penalties on lateral connections and sparsity-inducing gates on vertical connections. This encourages direct, clear causal paths and prunes edges whose influence can already be explained by multi-hop paths through other units.

Mathematically, do-influence can be expressed in terms of second derivatives of the free energy, or approximated via natural-gradient / Fisher-metric surrogates. In multilayer hierarchies, block Schur complements and Gauss-Newton approximations give tractable formulas for interventional maps from one layer to another.

The net effect is that CC drives the network toward a representation in which:

- causal influence flows along a relatively sparse backbone of strong edges;
- many potential paths exist but are weak or pruned;
- the effective mixed Hessian blocks between causally unrelated groups of parameters are small.

2.3 PC Transformers

Recent work has shown how to implement Transformers in a predictive-coding framework: self-attention, feed-forward layers, and normalization are treated as parts of a generative model; inference iteratively relaxes “value” nodes to minimize a KL-based free energy, and learning updates weights accordingly. The key take-away for our purposes is that:

- attention heads, value channels, and feed-forward blocks provide natural *modules* that could, in principle, be treated as approximately independent causal factors most of the time, with explicit couplings only where attention actually needs to connect tokens;
- predictive coding already introduces local error signals and recurrent inference that make it natural to implement causal gates and pruning at the head/channel level.

A “causal-codingified” PC Transformer would thus:

- estimate interventional influence from heads/channels in layer $\ell + 1$ to activities in layer ℓ ;
- gate updates to attention and feed-forward weights by these influence estimates;
- prune attention links and value channels that are redundant with others, using e.g. diffusion-style clarity penalties or sparse regression.

Section 7 returns to this in more detail.

2.4 Continual learning and commutativity

In continual learning, we are given a non-stationary stream of data in which the underlying distribution and objectives drift over time. We refer to the different regimes of this stream as *contexts*: different domains, goals, user populations, interaction styles, or environmental conditions. Catastrophic forgetting occurs when adapting to later contexts significantly degrades performance in earlier ones.

A useful way to analyze this is to view each context t as inducing a gradient vector field $X_t(\theta) = -\nabla L_t(\theta)$ on parameter space, where L_t is the loss under context t (e.g. an expectation over data from that context). Training for a small time h_t under context t is then approximated by the time- h_t flow $\Psi_{h_t}^t$ of X_t . A single pass over a sequence of contexts corresponds to the composition

$$\Theta_{\text{seq}} = \Psi_{h_T}^T \circ \dots \circ \Psi_{h_2}^2 \circ \Psi_{h_1}^1(\theta_0).$$

The “ideal” joint step, training on a weighted mixture $L_\Sigma = \sum_t h_t L_t$, corresponds to the time-1 flow of the combined vector field $X_\Sigma = \sum_t h_t X_t$:

$$\Theta_{\text{joint}} = \Psi_1^\Sigma(\theta_0).$$

General results on compositions of flows show that the difference between Θ_{seq} and Θ_{joint} is governed by the pairwise *commutators* (Lie brackets) $[X_i, X_j]$. Intuitively:

$$[X_i, X_j](\theta) = DX_j(\theta)X_i(\theta) - DX_i(\theta)X_j(\theta)$$

measures how much doing a tiny step of X_i then X_j differs from doing them in the opposite order. When all $[X_i, X_j]$ vanish, the flows *commute* and order does not matter. When $[X_i, X_j]$ are small, the order-dependence and hence the forgetting induced by a new context are quantitatively bounded.

For gradient flows, the commutator can be expressed in terms of mixed Hessian blocks, so that small cross-block Hessians (and/or small gradients) imply small commutators. This is where causal coding enters: by suppressing cross-module couplings, it shrinks the mixed blocks and thus reduces commutators across contexts. As we will see later via the categorical approximate CCL theorem, these effects can be quantified in terms of a sparse graph of “confusing” context pairs and do not require exact modularity.

3 The General Causal-Continual-Learning Theorem

We now formulate an abstract version of the main theorem, with as few architecture-specific assumptions as possible. This section uses a continuous notation (vector fields, gradients) but is really about the *structure* of context-conditioned updates; Section 5 gives a purely discrete version, and Appendix A gives a more general categorical version that subsumes both the continuous and discrete cases among other potential variations, and also provides approximate variants whose error bounds depend explicitly on context overlaps.

3.1 Abstract setup

Let Θ be a smooth parameter manifold (or simply $\mathbb{R}^d$). A *context* t is represented at the learning level by a smooth loss $L_t : \Theta \rightarrow \mathbb{R}$ with associated gradient vector field $X_t(\theta) = -\nabla L_t(\theta)$. We assume each L_t has Lipschitz gradient and (optionally) satisfies a Polyak-Lojasiewicz condition in the region of interest, so that gradient descent behaves well.

We suppose the learner has chosen a *causal representation and modularization* of parameters:

$$\theta = (\theta^{(1)}, \dots, \theta^{(M)}), \quad \theta^{(m)} \in \mathbb{R}^{d_m}, \quad \sum_m d_m = d,$$

intended to correspond to causal modules (latent factors, heads, channels, blocks, etc.). Informally, a given context t should depend strongly only on the subset of modules that causally matter for that context.

Definition 1 (Support sets and residual couplings). *For each context t we define:*

- a support set $S_t \subseteq \{1, \dots, M\}$ of modules that t claims to depend on;
- a residual gradient outside support

$$G_t^{out}(\theta) := \sum_{m \notin S_t} \left\| \frac{\partial L_t}{\partial \theta^{(m)}}(\theta) \right\|;$$

- cross-Hessian norms *between* modules $m \in S_t$ and $n \notin S_t$,

$$H_{t,m,n}^{cross}(\theta) := \left\| \frac{\partial^2 L_t}{\partial \theta^{(m)} \partial \theta^{(n)}}(\theta) \right\|.$$

We say that the family $\{L_t\}$ is $(\varepsilon_g, \varepsilon_h)$ -causally modular (relative to the modularization) if, in the training region,

$$G_t^{out}(\theta) \leq \varepsilon_g, \quad H_{t,m,n}^{cross}(\theta) \leq \varepsilon_h$$

for all t , all θ , all $m \in S_t$, and all $n \notin S_t$.

Intuitively: each context’s gradient lives almost entirely inside its support modules, and the curvature coupling its support modules to others is small. This definition controls the “outside-support” and cross-support terms; in the presence of overlapping supports, there are also in-support terms that need to be bounded separately in terms of a pairwise overlap or confusion score, as made explicit in the approximate categorical formalism in Appendix A.

In a causal-coding network, the support sets S_t and small residuals are enforced by causal gates and pruning:

- gates suppress updates on parameters whose do-influence under that context is tiny;
- clarity penalties prune edges that merely create redundant diffusive paths, shrinking cross-Hessian blocks.

3.2 From causal modularity to small commutators

Define the gradient vector fields

$$X_t(\theta) = -\nabla L_t(\theta) = -\left(\frac{\partial L_t}{\partial \theta^{(1)}}, \dots, \frac{\partial L_t}{\partial \theta^{(M)}}\right).$$

The Lie bracket $[X_i, X_j]$ can be expressed in components as

$$[X_i, X_j]^k(\theta) = \sum_{\ell} \left(\frac{\partial X_j^k}{\partial \theta^\ell} X_i^\ell - \frac{\partial X_i^k}{\partial \theta^\ell} X_j^\ell \right),$$

where k, ℓ range over coordinates on Θ . In block form, one can think of this as involving mixed Hessian blocks of L_i and L_j weighted by gradient components of L_j and L_i , respectively.

The following lemma abstracts the usual “commutator is bounded by mixed Hessians times gradients” calculation to the modular setting and captures those parts of the commutator controlled by out-of-support gradients and cross-support Hessians. Additional in-support terms arising from overlapping supports are handled via pairwise confusion costs in the approximate categorical theorem in Appendix A.

Lemma 1 (Causal modularity implies small commutator contributions). *Suppose the family $\{L_t\}$ is $(\varepsilon_g, \varepsilon_h)$ -causally modular, with all ∇L_t uniformly bounded by $G_{\max}$ in norm. Then there exists a constant C depending only on the modularization such that, for all $i \neq j$ and all θ in the training region, the contribution to $[X_i, X_j](\theta)$ coming from cross-support Hessian blocks and out-of-support gradients obeys*

$$\|[X_i, X_j](\theta)\|_{\text{cross}} \leq C (\varepsilon_h G_{\max} + \varepsilon_g H_{\max}),$$

where $H_{\max}$ bounds the within-support Hessian blocks in norm. In the special case where supports are disjoint, this bound controls the full commutator.

Sketch. Write the gradient and Hessian in block form with respect to the modules. Within each context t , blocks linking modules outside S_t to those inside S_t are bounded by ε_h , and gradient components outside S_t are bounded by ε_g . Expanding $[X_i, X_j]$ in block form shows that every term that involves a cross-support Hessian block or an out-of-support gradient either contains a cross-Hessian block (bounded by ε_h) times a gradient (bounded by $G_{\max}$), or a within-support Hessian block (bounded by $H_{\max}$) times an out-of-support gradient (bounded by ε_g). Summing over finitely many modules produces the stated bound with some combinatorial constant C . When supports are disjoint, there are no in-support overlap terms and this bound controls the entire bracket. $\square$

Thus, in a causally modular representation, a large part of the commutators of context vector fields is small by construction, and the remaining contributions arise from overlaps in the supports $S_i \cap S_j$; these overlap terms are controlled in Appendix A by *confusion costs* that quantify how strongly two contexts interact on their shared modules.

3.3 Sequential vs. joint training under small commutators

Let Ψ_h^t denote the time- h flow of X_t , and write a single training cycle as before:

$$\Theta_{\text{seq}} = \Psi_{h_T}^T \circ \dots \circ \Psi_{h_1}^1(\theta_0), \quad \Theta_{\text{joint}} = \Psi_1^\Sigma(\theta_0),$$

with $X_\Sigma = \sum_t h_t X_t$.

General results on compositions of flows (via Taylor expansion or the Baker-Campbell-Hausdorff formula) give

$$\Theta_{\text{seq}} - \Theta_{\text{joint}} = \frac{1}{2} \sum_{i < j} h_i h_j [X_i, X_j](\theta_0) + O\left(\sum_{i,j,k} h_i h_j h_k\right),$$

and hence, under a uniform bound $\|[X_i, X_j](\theta)\| \leq \varepsilon$,

$$\|\Theta_{\text{seq}} - \Theta_{\text{joint}}\| \leq \frac{1}{2} \varepsilon \sum_{i < j} h_i h_j + O(h^3),$$

where $h = \max_t h_t$.

Moreover, the change in an earlier context's loss L_i after training under a later context j can be bounded in terms of $[X_i, X_j]$ and Lipschitz constants of ∇L_i . Informally, the more the new context's flow fails to commute with the old context's flow, the more it pulls parameters off the old context's solution manifold. In the approximate categorical setting this is made precise by bounds that depend only on the confusion costs of those pairs (i, j) that actually interfere.

3.4 The General Causal-Continual-Learning Theorem

We can now combine causal modularity and flow composition to state the main theorem.

Theorem 1 (General Causal-Continual-Learning Theorem). *Let Θ be a smooth parameter manifold and $\{L_t\}_{t=1}^T$ a family of C^3 context-conditioned losses with Lipschitz gradients and bounded Hessians on a convex training region $N \subset \Theta$. Suppose:*

1. (Causal modularity) *There is a modularization $\theta = (\theta^{(1)}, \dots, \theta^{(M)})$ and support sets S_t such that the family $\{L_t\}$ is $(\varepsilon_g, \varepsilon_h)$ -causally modular on N , with uniform bounds $G_{\max}$ and $H_{\max}$ on gradient norms and within-support Hessian blocks.*
2. (Causal coding realizer) *The learning algorithm in each context approximately follows gradient flow on L_t with respect to this modularization, using causal gates and pruning derived from (exact or approximate) do-influence. That is, the actual update vector field $\tilde{X}_t$ satisfies $\|\tilde{X}_t - X_t\| \leq \delta$ on N , and the gates are computed from influence estimates that agree (up to δ) with the true causal influence.*
3. (Step sizes) *Step sizes $h_t > 0$ are small enough that all flows stay within N and higher-order error terms $O(h^3)$ can be neglected.*

Then there exists a constant C such that, for any initial $\theta_0 \in N$:

1. (Approximate commutativity) *For all $i \neq j$ and all $\theta \in N$,*

$$\|[\tilde{X}_i, \tilde{X}_j](\theta)\| \leq C(\varepsilon_h G_{\max} + \varepsilon_g H_{\max} + \delta) := \varepsilon_{\text{comm}},$$

where $\varepsilon_{\text{comm}}$ should be understood as a worst-case bound that can be refined in terms of pairwise confusion costs as in the approximate categorical theorem.

2. (Sequential $\approx$ joint) *Let $\tilde{\Psi}_{h_t}^t$ be the time- h_t flow of $\tilde{X}_t$, and $\tilde{\Psi}_1^\Sigma$ the time-1 flow of $\tilde{X}_\Sigma := \sum_t h_t \tilde{X}_t$. Then*

$$\|\tilde{\Theta}_{\text{seq}} - \tilde{\Theta}_{\text{joint}}\| \leq \frac{1}{2} \varepsilon_{\text{comm}} \sum_{i < j} h_i h_j + O(h^3),$$

where $\tilde{\Theta}_{\text{seq}} = \tilde{\Psi}_{h_T}^T \circ \dots \circ \tilde{\Psi}_{h_1}^1(\theta_0)$ and $\tilde{\Theta}_{\text{joint}} = \tilde{\Psi}_1^\Sigma(\theta_0)$.

3. (Forgetting bound) For any earlier context i and later context j , the change in L_i after a step on context j satisfies

$$|L_i(\theta^{\text{after } j}) - L_i(\theta^{\text{before } j})| \leq K_{ij} h_i h_j \varepsilon_{\text{comm}} + O(h^3),$$

where K_{ij} depends on Lipschitz constants of ∇L_i along the trajectory. In particular, catastrophic forgetting is ruled out in the limit $\varepsilon_{\text{comm}} \rightarrow 0$ and small h_t , and more refined bounds that depend only on confusing pairs are given by the approximate categorical theorem in Appendix A.

Sketch. Item (1) follows by combining Lemma 1 (applied to X_t) with the fact that replacing X_t by $\tilde{X}_t$ perturbs commutators by at most $O(\delta)$. Overlapping-support contributions are further controlled by pairwise confusion costs as in the enriched categorical setting.

Item (2) is a direct application of the standard expansion for compositions of flows: the deviation between the product of flows and the flow of the sum is given to second order by a sum of pairwise commutators weighted by $h_i h_j / 2$, plus higher-order nested commutators which are $O(h^3)$ under our smoothness assumptions.

Item (3) applies the mean value theorem to L_i along the segment between $\tilde{\Theta}_{\text{seq}}$ and $\tilde{\Theta}_{\text{joint}}$, plus the bound from (2), to relate the change in L_i to $\varepsilon_{\text{comm}}$ and Lipschitz constants of ∇L_i . Details are standard and omitted. $\square$

3.5 Informal interpretation

The theorem formalizes the philosophical story from the Introduction:

- If the learner has found a representation in which each context touches only the modules it really needs (causal modularity), and the residual couplings between modules are small, then the context gradient fields almost commute. Causal coding is one concrete way to enforce this modularity.
- When context flows almost commute, the difference between seeing contexts in sequence and seeing them jointly is a second-order effect controlled by pairwise commutators. In particular, later contexts cannot drastically degrade earlier ones unless the commutators are large, and the approximate categorical theorem refines this to show that only “confusing” pairs matter.
- The independence/coupling tradeoff theorem tells you *why* the learner would want to be in the causal-modular regime: there, it achieves a good accuracy/compute tradeoff by keeping modules as independent as the data allow and only paying for explicit couplings where necessary.
- Thus causal modeling (to identify the right modules) plus a drive to maximally independent representations (to sit on the Pareto frontier of the independence/coupling tradeoff) yields a regime in which continual learning is automatically well-behaved.

In short: if you build and maintain the right causal factorization, the commutators shrink, and so does forgetting, with the quality of continual learning degrading smoothly as modularity assumptions are relaxed.

3.6 Making the theorem PAC for closer correlation to reality

Limitations and approximate categorical variants. Theorem 1, as stated above, should be read as a clean, worst-case continuous instantiation of the general idea rather than as the final word. Its hypotheses pack all violations of modularity and commutativity into a single uniform parameter $\varepsilon_{\text{comm}}$, and the resulting bounds use this worst-case constant for every pair of contexts. In practice, however, neither exact modularity nor uniform commutator bounds are realistic: contexts typically have overlapping supports and some pairs of contexts interact much more strongly than others. The continuous theorem also does not explicitly separate the different sources of non-commutativity (overlaps in $S_i \cap S_j$ versus approximation errors in the update rule), but rather folds them into a single quantity.

To address this, Appendix A develops an *approximate categorical* version of the CCL theorem in a metric-enriched monoidal setting. There we make explicit two kinds of error: (i) a *locality error* ε_c for each context c , measuring how far its update U_c is from an ideal context-local map that acts only on its intended support S_c , and (ii) an *ideal commutator defect* $\Delta_{c,c'}$ for each pair of contexts, measuring how far their ideal updates fail to commute. These combine into a per-pair *confusion cost* $\delta_{c,c'}$, and the main approximate categorical theorem shows that path-dependence and forgetting error can be bounded by a sum over these confusion costs. In particular, the deviation between sequential and joint updates is shown to scale linearly in the number and magnitude of genuinely confusing context pairs, rather than in the square of the total number of contexts.

The continuous and discrete specializations in the main text arise as instances of this enriched categorical picture: in the smooth case we take Θ to be a parameter manifold, U_c to be small-gradient flows for L_c , and d_Θ to measure performance distance; in the discrete case we take Θ to be a configuration space with a metric d and U_c to be non-expansive update maps. In both cases the categorical PAC-style formulation replaces the single, global $\varepsilon_{\text{comm}}$ of Theorem 1 by a structured family of errors $\{\varepsilon_c, \Delta_{c,c'}\}$ and makes clear how continual-learning guarantees degrade smoothly as modularity assumptions are relaxed. This refined perspective is important for practical applications: real systems are only approximately modular, and it is the sparsity and weakness of the confusion graph (rather than exact disjointness) that determines how robust sequential training will be.

4 Continuous-variable specialization

The previous section was already stated in a smooth setting. Here we make the gradient-flow assumptions explicit and connect more directly to standard continuous-time analysis.

4.1 Continuous parameter manifolds and gradient flows

Assume $\Theta \subseteq \mathbb{R}^d$ is open and each L_t is C^3 . Define the gradient vector fields $X_t = -\nabla L_t$ and flows Ψ_h^t as before. We assume:

- *Lipschitz gradient*: $\|\nabla L_t(\theta) - \nabla L_t(\theta')\| \leq L_t \|\theta - \theta'\|$ for all θ, θ' in a convex region N , with uniform bound $L_t \leq L$.
- *Bounded Hessian*: $\|\nabla^2 L_t(\theta)\| \leq H$ on N .
- *PL inequality (optional)*: $\frac{1}{2} \|\nabla L_t(\theta)\|^2 \geq \mu(L_t(\theta) - L_t^*)$, ensuring gradient descent converges linearly to a minimum.

Under these assumptions, classical ODE theory justifies the Taylor expansions of flows and the Baker-Campbell-Hausdorff truncations used in Theorem 1.

4.2 Explicit Lie-bracket bounds

In coordinates, the Lie bracket for gradient fields can be written as

$$[X_i, X_j](\theta) = -\nabla^2 L_j(\theta) \nabla L_i(\theta) + \nabla^2 L_i(\theta) \nabla L_j(\theta).$$

In block form with respect to a parameter split $\theta = (u, v)$, this yields

$$[X_i, X_j](\theta) = \begin{pmatrix} -H_{uu}^{(j)} g_u^{(i)} - H_{uv}^{(j)} g_v^{(i)} + H_{uu}^{(i)} g_u^{(j)} + H_{uv}^{(i)} g_v^{(j)} \\ -H_{vu}^{(j)} g_u^{(i)} - H_{vv}^{(j)} g_v^{(i)} + H_{vu}^{(i)} g_u^{(j)} + H_{vv}^{(i)} g_v^{(j)} \end{pmatrix},$$

where $H_{\cdot\cdot}^{(t)}$ are Hessian blocks of L_t and $g^{(t)}$ the corresponding gradient blocks.

If causal coding and pruning have suppressed the mixed blocks H_{uv} and H_{vu} , then the commutator norm is dominated by within-block Hessians times cross-block gradients—exactly the pattern captured in Lemma 1. Quantitatively, if $\|H_{uv}^{(t)}\|, \|H_{vu}^{(t)}\| \leq \varepsilon_h$ and $\|g_v^{(t)}\| \leq \varepsilon_g$, then

$$\|[X_i, X_j](\theta)\| \leq 2H\varepsilon_g + 2G_{\max}\varepsilon_h,$$

with H a bound on within-block Hessians and $G_{\max}$ a bound on gradient norms. Overlaps in the supports $S_i \cap S_j$ contribute additional terms that can be bounded in terms of confusion costs $\delta_{i,j}$ as in Appendix A.

Thus the continuous specialization of Theorem 1 is essentially Theorem 1 itself, with explicit continuous-time constants and with more refined, overlap-sensitive versions available via the categorical approximate CCL theorem.

5 Discrete-variable specialization

We now give a version that does not assume a differentiable structure at all. This captures, for example, symbolic program rewriting systems, finite-state controllers, or modular networks updated by discrete rules.

5.1 Abstract endomorphisms and commutators

Let Θ be an arbitrary set of states or configurations. A context t is represented at the update level by an endomorphism $U_t : \Theta \rightarrow \Theta$ (e.g. one training epoch or one adaptation step under that context). Sequential updates across a sequence of contexts $1, \dots, T$ give

$$\Theta_{\text{seq}} = U_T \circ \dots \circ U_1(\theta_0).$$

We can also consider a “joint” update U_Σ that approximates the effect of training on some combination of contexts (e.g. stochastic mixtures of minibatches from different parts of the data stream).

To talk about “small” deviations we equip Θ with a (pseudo)metric $d : \Theta \times \Theta \rightarrow [0, \infty)$, or more generally with an enrichment in a quantale of costs/effects. For simplicity assume a metric.

Define the *discrete commutator* of two updates U_i, U_j at a point θ by

$$C_{ij}(\theta) := U_j(U_i(\theta)), \quad C_{ji}(\theta) := U_i(U_j(\theta)).$$

We say U_i and U_j commute up to ε if $d(C_{ij}(\theta), C_{ji}(\theta)) \leq \varepsilon$ for all θ .

5.2 Discrete causal modularity

Suppose we have a decomposition of state into components $\theta = (\theta^{(1)}, \dots, \theta^{(M)})$ and a notion of projection $\pi^{(m)} : \Theta \rightarrow \Theta^{(m)}$ (for instance Θ is a product space). For each context t we define a support set S_t and say that U_t is *locally causal* if it changes components in S_t arbitrarily but affects components outside S_t only through small, bounded interactions.

One way to formalize this is:

Definition 2 (Discrete causal modularity). *We say a family $\{U_t\}$ is ε -causally modular if there exists a decomposition $\Theta \cong \prod_m \Theta^{(m)}$ and support sets S_t such that, for all θ and all $m \notin S_t$,*

$$d(\pi^{(m)}(U_t(\theta)), \pi^{(m)}(\theta)) \leq \varepsilon.$$

That is, each context-conditioned update U_t leaves non-support modules almost fixed.

In a causal-coding implementation, this arises from gates that freeze modules with negligible do-influence for the current context, and from pruning that removes cross-module couplings. The metric d can capture, for example, differences in loss, in causal influence structure, or in a summary of the state.

5.3 Discrete Causal-Continual-Learning Theorem

Theorem 2 (Discrete Causal-Continual-Learning). *Let Θ be a metric space with metric d , and let $\{U_t\}_{t=1}^T$ be a family of endomorphisms that is ε -causally modular with respect to some decomposition and support sets S_t . Suppose further that each U_t is non-expansive in d (Lipschitz constant ≤ 1).*

Then there exists K depending only on the decomposition such that, for all $i \neq j$ and all θ ,

$$d(U_j(U_i(\theta)), U_i(U_j(\theta))) \leq K\varepsilon.$$

In particular, the deviation between sequential and joint updates after one cycle is bounded by $O(T^2\varepsilon)$ in d , and the change in any context-specific evaluation functional that is Lipschitz in d is bounded by $O(T^2\varepsilon)$ as well. Appendix A shows how this worst-case $O(T^2)$ scaling can be refined to bounds that depend linearly on the number and magnitude of pairwise confusion costs $\delta_{i,j}$, and hence on the number of genuinely confusing context pairs rather than all T^2 pairs.

Sketch. Because each U_t leaves non-support modules almost fixed, composing U_i and U_j in different orders can only differ to the extent that their supports overlap and propagate into non-support modules. The number of overlapping modules and the depth of such propagation is bounded by a constant depending on the decomposition. Each application introduces at most ε disturbance per non-support module, and non-expansiveness prevents amplification. Thus the total divergence between $U_j U_i$ and $U_i U_j$ is at most $K\varepsilon$ for some constant K . A union bound over pairs and steps then gives the $O(T^2\varepsilon)$ bound for a full cycle. $\square$

This theorem is the discrete analog of Theorem 1: causal modularity (in the sense of almost disjoint supports) implies small commutators and hence bounded forgetting, without any differentiability assumption. The approximate categorical CCL theorem in Appendix A shows how to sharpen this to bounds that depend linearly on the number and strength of confusing context pairs.

5.4 Categorical viewpoint (very briefly)

Both the continuous and discrete versions can be seen as instances of a more general categorical statement: in a symmetric monoidal category $\mathcal{C}$ enriched in a quantale Q , with an object decomposed as a tensor product of causal subsystems, a family of endomorphisms that each act as

identity on all but their support factors, and whose nontrivial actions are Q -small outside support, will compose in an almost-commutative way with respect to the Q -order. The metric d or the norm on vector fields is then just one choice of such an enrichment.

The details of the categorical formulation, including approximate variants with confusion costs, are given in Appendix A at the end of the paper.

6 Application to causal-coding neural networks

We now specialize the continuous theorem to causal-coding predictive networks.

6.1 Parameter modularization

In a typical PC/CC decoder we can take the parameter vector to be grouped into:

- vertical weights $W^{(\ell)}$ between layers;
- lateral weights $L^{(\ell)}$ within layers;
- possibly biases and normalization parameters.

A natural modularization is to treat each vertical or lateral weight block (as a matrix, convolutional kernel, head, or channel) as a module $\theta^{(m)}$.

6.2 Causal gates and clarity as enforcers of modularity

Causal coding estimates do-influence matrices C layerwise, either exactly (via Hessians) or approximately (via Fisher surrogates or learned approximators). It then:

- constructs gates $r_{ji} \propto |C_{ji}|^p$ to modulate weight updates, suppressing updates on edges with negligible causal influence;
- applies clarity or diffusion-based penalties that shrink or prune lateral edges whose effect can be reproduced by multi-hop paths.

These mechanisms directly shrink:

- gradients of L_t with respect to parameters that have negligible do-influence in context t (reducing ε_g);
- effective mixed Hessian blocks between modules that do not share strong causal paths (reducing ε_h).

Thus, under sustained causal coding, the network is driven into a regime where the loss family $\{L_t\}$ associated with different contexts is $(\varepsilon_g, \varepsilon_h)$ -causally modular in the sense of Definition 1, and where pairwise confusion costs between contexts are small for all but a sparse set of genuinely interacting context pairs.

6.3 Resulting continual-learning guarantees

Given the above, Theorem 1 applies with:

- modules corresponding to vertical and lateral weight blocks, attention heads, channels, etc.;
- support sets S_t corresponding to those modules whose gates are non-negligible in context t ;
- $\varepsilon_g, \varepsilon_h$ controlled by gate selectivity, pruning thresholds, and the amount of variance in do-influence estimates.

If causal coding is aggressive enough (strong sparsity, strong clarity penalties) to keep $\varepsilon_{\text{comm}}$ and the confusion costs small, then:

- one pass through a stream of contexts is close (in parameter space and in function space) to a joint step on the corresponding mixture loss;
- the degradation in performance under earlier contexts when adapting to new ones is bounded by a term proportional to $\varepsilon_{\text{comm}}$ and to the number of confusing context pairs encountered;
- catastrophic forgetting can only occur when contexts genuinely conflict at the causal level (i.e. when they require incompatible assignments to the same causal modules).

In this way, causal coding provides a principled mechanism to achieve the structural conditions (causal modularity, small commutators, sparse confusion graph) that make continual learning robust.

7 Application to PC Transformers

Next we sketch how the approach we are exploring applies to predictive-coding Transformers augmented with causal coding.

7.1 Modularization in a Transformer

In a Transformer block we can distinguish several natural modules:

- self-attention heads $(W_Q^{(h)}, W_K^{(h)}, W_V^{(h)}, W_O^{(h)})$ for each head h ;
- feed-forward channels (rows of the feed-forward matrices);
- layer-norm and positional-encoding parameters.

A PC Transformer introduces additional “activity” parameters for each layer (the PC value nodes) and inference dynamics over them, but the weight parameters remain grouped in the same way.

7.2 Causal coding at the head/channel level

A causal-coding PC Transformer would:

- estimate interventional influence of each head/channel on downstream prediction errors (e.g. how much the equilibrium distribution at some layer changes under do-interventions on that head);
- build gates per head (and possibly per token) based on these influences;
- apply clarity penalties to prune attention links and channels that are redundant with others (e.g. via diffusion on the token/head graph).

Intuitively, most of the time many heads are not causally necessary for a given input and surrounding context; gates can shut them off, making their parameters modules outside the support set S_t for that context. Clarity penalties discourage diffuse, multi-hop attention patterns when direct edges suffice.

7.3 Causal modularity and commutators for Transformer contexts

In this setting, contexts might be different input domains, prompt styles, interaction regimes, or implicit tasks defined by user intent. Each context t induces a loss L_t on the Transformer parameters. Under a causal-coding regime:

- For a given context, only a subset of heads and channels (support modules) receive strong gradient updates; gradients on other modules are suppressed by gates (small ε_g).
- Mixed Hessian blocks coupling support modules for one context to support modules unique to another context shrink as the representation is refactorized and the network learns to route context-specific information along dedicated paths (small ε_h), and confusion costs are concentrated on a sparse set of overlapping contexts.

Thus, the family $\{L_t\}$ becomes causally modular with respect to the head/channel decomposition, satisfying the hypotheses of Theorem 1 and the approximate categorical CCL theorem. Consequently:

- The gradient flows for different contexts approximately commute; training under one contextual distribution and then another is close to training on the mixture, with a deviation that grows linearly in the number of confusing context pairs.
- Catastrophic forgetting is bounded: behavior in “old” contexts is only harmed by “new” contexts to the extent that they really share and conflict over the same heads and channels.
- The independence/coupling tradeoff provides design guidance: invest Route A effort in finding a head/channel factorization that makes contexts as independent as possible, and Route B effort in explicit cross-token attention only where causal evidence demands it.

This suggests a principled route to continual learning in Transformer-like architectures: use predictive coding to get local error signals and biologically-plausible inference, use causal coding to discover and enforce a modular causal factorization over heads and channels, and rely on the general and approximate Causal-Continual-Learning Theorems to guarantee that, once this factorization

is in place, sequential adaptation across a stream of contexts will behave similarly to joint training and will not catastrophically forget.

Appendix B fleshes out this proposed application further, elaborating a hypothetical concrete example of how a causal coding based transformer might deal with prompts spanning different contexts.

8 Application to incremental compression as a learning approach

Slightly more adventurously, we now explore how the general relationship between causal coding and continual learning might manifest itself in the context of a different (non-neural) sort of learning mechanism: incremental compression a la the WILLIAM algorithm. The goal here is not to claim that arbitrary incremental compressors will automatically become modular, but rather to propose a class of *causal IC* update rules that are designed to satisfy the CCL hypotheses and to illustrate how the general theory informs the design.

8.1 Incremental-compression-style causal coding and continual learning

Informally, the difference between ordinary WILLIAM-style incremental compression and the causal-incremental version here is about how much the system remembers *which* patterns were learned *for which contexts*, and how it *adapts* its bias toward sharing structure across contexts versus keeping it separate.

In plain incremental compression (as in WILLIAM), there is a single global feature library. Whenever new data arrives, the system just tries to improve overall compression on that data: it adds new features, refits old ones, and may prune features that no longer help. If the stream of data is a mixture of many contexts, then from the compressor’s point of view there is just one big distribution. It has no explicit reason to protect features that were important for some earlier context but are not helpful for the current batch, nor to regulate how aggressively features are shared across contexts.

Depending on hyperparameters, this can fail in two opposite ways:

- If refitting and pruning are aggressive, then when a new context arrives the search procedure will happily reshape or delete features that used to explain an old context well. Compression on the new context improves, but compression on the old one can get much worse. This is a form of catastrophic forgetting.
- If we make the system very conservative (strong regularization, weak refitting, strict pruning thresholds) to avoid forgetting, then the global library becomes biased toward the earliest or most frequent contexts. New contexts are forced to reuse old features even when they are not a good fit, so adaptation to new contexts becomes slow and brittle.

The causal-incremental scheme described below tries to thread the needle. We still use incremental compression and refitting, but we:

- organize the feature library into modules;
- measure, per context, how much each module actually helps compression (a kind of do-influence);
- update only the modules that really matter for the current context, freezing the others;

- add a clarity penalty so that features do not try to simultaneously explain many unrelated contexts, and *meta-learn* the strength and shape of this penalty from experience so that the system automatically chooses how much sharing versus separation is appropriate for a given environment.

In the ideal limit, each context mostly updates its own subset of modules, shared modules only carry genuinely shared structure, and the meta-learned clarity penalties settle into a regime where useful sharing is kept and spurious overlap is discouraged. Updates for different contexts then almost commute, so training on contexts in different orders gives almost the same library and the same compression performance. This is exactly the situation captured by the discrete/categorical Causal-Continual-Learning theorems: context updates are almost factorwise endomorphisms, so sequential multi-context learning has bounded forgetting. The construction below should be read as a *design pattern* that realizes these hypotheses for incremental compression, not as a claim that arbitrary IC schemes will converge to a modular representation on their own.

8.2 Contexts, feature libraries, and modules

Fix string length N and a family of *contexts* $c \in \mathcal{C}$, each given by a distribution P_c over $\{0, 1\}^N$. For a given context c and string $x \sim P_c$, an incremental decomposition is a sequence of features $F = (f_1, \dots, f_s)$ and residual r_s as in the incremental compression framework: a feature is an invertible, length-preserving program f with description length $K(f)$, residuals evolve via $r_i = f_i(r_{i-1})$, and the total two-part code length is

$$L_s(F; x) := \sum_{i=1}^s K(f_i) + K(r_s),$$

with efficiency

$$\eta(F; x) := \frac{N - L_s(F; x)}{N - K(x)} \in [0, 1].$$

The learner does not start from scratch for each context; instead it maintains a *global feature library* and reuses it. Formally, the internal state is

$$\Theta = (F^{(1)}, \dots, F^{(K)}),$$

where each $F^{(k)}$ is a finite set of short features and the K blocks are treated as *modules* or *causal factors*. Think of $F^{(1)}$ as “low-level image patches”, $F^{(2)}$ as “layout templates”, $F^{(3)}$ as “text overlays”, and so on, but in a purely algorithmic/incremental-coding sense.

In addition, we will allow each module to carry a small collection of *meta-parameters* controlling clarity and sharing, for example per-feature or per-module clarity weights. For notational simplicity we continue to write Θ for the whole library state, understanding that these meta-parameters live inside the modules alongside the features.

Given Θ and a context c , the compressor is allowed to use features from all modules, but we will explicitly track how much each module matters. Let $L_c(\Theta)$ denote the expected two-part description length (over P_c) when we run some fixed incremental-coding search (for example greedy-with-refit or PGM-guided search) using only features from Θ . We define a simple metric on states:

$$d(\Theta, \Theta') := \sup_{c \in \mathcal{C}} |L_c(\Theta) - L_c(\Theta')|.$$

Context supports and do-influence. For each context c and module k , define a *do-influence* number by the counterfactual compression gap

$$I(c, k; \Theta) := L_c(\Theta_{\setminus k}) - L_c(\Theta),$$

where $\Theta_{\setminus k}$ is obtained from Θ by disabling all features in $F^{(k)}$ (we run the same incremental search but are not allowed to use those features). Intuitively, $I(c, k; \Theta)$ is how many bits we lose on context c if we intervene to delete module k .

Given a threshold $\tau > 0$, we define the *support* of context c at state Θ as

$$S_c(\Theta) := \{k : I(c, k; \Theta) \geq \tau\}.$$

Modules outside $S_c(\Theta)$ are (at this scale) causally irrelevant for compressing c .

8.3 An incremental-causal update rule for a new context

We now describe a “causal coding esque” update rule $U_c : \Theta \rightarrow \Theta$ for training on a batch of data from context c .

Given current library Θ and a finite sample $D_c = \{x^{(1)}, \dots, x^{(m)}\}$ from P_c , one update step for context c proceeds as follows.

1. **Estimate do-influences and supports.** Approximate $L_c(\Theta)$ and $L_c(\Theta_{\setminus k})$ by running the incremental compressor on D_c with and without module k , and compute $I(c, k; \Theta)$. Form the support

$$S_c := S_c(\Theta) = \{k : I(c, k; \Theta) \geq \tau\}$$

for a chosen threshold τ . Optionally define soft gates $r_{c,k} := \psi(I(c, k; \Theta)) \in [0, 1]$ with some monotone schedule ψ .

2. **Context-local feature refinement (within support).** For each $k \in S_c$, run a one- or few-step incremental-compression search restricted to features in $F^{(k)}$ plus candidate new features proposed from the residuals on D_c , using some algorithm such as greedy-with-refit, beam search, or PGM-guided search. Within each module we:
 - add new short features f that achieve positive net gain on D_c ;
 - refit all features in $F^{(k)} \cup \{f\}$ jointly against D_c , so that they re-center on the structure specific to k (preventing earlier features from absorbing higher-level structure, as in the single-context refit analysis);
 - keep track, for each $f \in F^{(k)}$, of its per-context gain $\gamma_c(f)$ and cost $\tau(f)$.
3. **Clarity / de-overlap across contexts, with meta-learned strength.** To prevent modules from becoming entangled across contexts, we introduce a *clarity penalty* that discourages features from carrying redundant mass for many unrelated contexts. Concretely, maintain for each feature f the vector $\gamma_c(f)$ of per-context gains and define its cross-context overlap

$$\text{Overlap}(f) := \sum_{c \neq c'} \min\{\gamma_c(f), \gamma_{c'}(f)\}.$$

Rather than fixing a single hand-chosen penalty weight, we treat a family of clarity weights $\lambda_f \geq 0$ (or, more coarsely, λ_k per module) as part of the state Θ , and update them by a slow *meta-learning* rule.

During refitting for context c we minimize an objective of the schematic form

$$\tilde{L}_c(\Theta) := L_c(\Theta) + \sum_f \lambda_f \text{Overlap}(f),$$

and after processing several contexts we take meta-steps that adjust the λ_f to improve long-run multi-context performance, for example by approximately decreasing a global meta-loss

$$L_{\text{meta}}(\Theta, \lambda) := \mathbb{E}_{c \sim \pi} [L_c(\Theta)] + \kappa \sum_f \lambda_f \text{Overlap}(f)$$

for some sampling distribution π over contexts and tradeoff parameter κ . Intuitively, the meta-learner increases λ_f when a feature’s overlap looks like unhelpful “diffuse sharing” (small extra compression in many contexts that could be better explained by more local features), and decreases λ_f when sharing truly improves compression on multiple related contexts.

Operationally, this encourages the learner to:

- keep truly shared structure in a dedicated “shared” module with appropriately small penalties;
- split features that mix unrelated contexts into separate, more local features;
- prune features that contribute only tiny extra compression on many contexts (that is, weak, diffusive features).

This is the analogue of clarity or diffusion penalties in causal coding, but now with the penalty strengths themselves adapted online so that the amount of enforced separation or sharing matches the environment.

4. **Freezing outside support.** For $k \notin S_c$, we freeze the module: no new features are added to $F^{(k)}$ and existing features in $F^{(k)}$ are not refit on D_c . In practice one can allow extremely small meta-updates (for example annealed smoothing of global statistics or of clarity weights), but we assume these are bounded by a small $\varepsilon_{\text{leak}}$ in the metric d defined above.

The net result is a map $U_c : \Theta \rightarrow \Theta'$ which only significantly modifies the modules in S_c , and does so using an incremental-compression objective that explicitly penalizes cross-context overlap with penalty strengths that are themselves meta-learned from experience. This U_c is by construction designed to satisfy the discrete CCL hypotheses with small leakage and overlap, making it a concrete instantiation of the general theory.

8.4 Structural assumptions for multi-context incremental compressibility

To connect this to the general CCL theorem, we need the analogue of “causal modularity” for incremental compression.

Assume that:

- Each context c is generated by a short-feature hierarchical generative model (SF-HGM) with k_c levels and bounded cross-information δ between levels, as in the incremental-compression analysis. This implies that strings from P_c admit k_c -incremental decompositions with high efficiency $\eta \geq 1 - O(\delta k_c)$, using short features with small overlaps.

- The contexts themselves are not maximally entangled: there is a factorization of the causal structure such that, after an appropriate change of variables, each context c uses only a subset S_c of the modules, and cross-context mutual information between modules is bounded (same spirit as bounded cross-information, now across contexts rather than across levels).
- The incremental search (greedy-with-refit, PGM-guided, and so on) is strong enough to recover near-optimal, low-overlap features for each context when restricted to its support modules, as in the single-context results; and the meta-learning rule for the clarity weights λ_f converges (in a coarse sense) to a regime where features with large, unhelpful cross-context overlap are either moved into explicitly shared modules or pruned, so that the residual “diffusive” overlap remains uniformly small.

Under these assumptions, we can choose the modules Θ_k so that each context’s best incremental decomposition lives almost entirely inside its support modules, and the meta-learned clarity penalties keep cross-context overlaps small, making the confusion graph sparse and the discrete CCL bounds tight.

8.5 Incremental-Compression Causal-Continual-Learning Theorem

We now state a discrete CCL theorem specialized to this setting.

Theorem 3 (Incremental-compression CCL). *Let Θ be a feature library factorized into modules $\Theta = (\Theta_1, \dots, \Theta_K)$, with metric*

$$d(\Theta, \Theta') = \sup_{c \in \mathcal{C}} |L_c(\Theta) - L_c(\Theta')|.$$

Let $\{U_c : \Theta \rightarrow \Theta\}_{c \in \mathcal{C}}$ be the context-update maps defined by the gated, refit-and-clarify rule above, including the slow meta-updates of the clarity weights.

Assume:

1. Context-locality. *For each context c there is a support $S_c \subseteq \{1, \dots, K\}$ such that, for all states Θ in the training region,*

$$d(U_c(\Theta), \Theta) \leq \varepsilon_{tot}, \quad d(\pi_k(U_c(\Theta)), \pi_k(\Theta)) \leq \varepsilon_{leak} \quad \text{for all } k \notin S_c,$$

where π_k projects Θ to its k th module and $\varepsilon_{leak} \ll \varepsilon_{tot}$ measures leakage outside the support.

2. Non-expansiveness. *Each U_c is 1-Lipschitz with respect to d , i.e.*

$$d(U_c(\Theta), U_c(\Theta')) \leq d(\Theta, \Theta') \quad \text{for all } \Theta, \Theta'.$$

3. Small cross-context overlap. *The combination of SF-HGM structure and the meta-learned clarity weights enforces that whenever $c \neq c'$ and $k \in S_c \setminus S_{c'}$, the incremental gain of context c' from features in module k is at most $\varepsilon_{overlap}$ in normalized units. Equivalently, after the meta-learner has had time to adapt, any feature that still carries large overlap for many unrelated contexts must live in a dedicated shared module whose contribution is accounted for separately, and the remaining “context-specific” modules are nearly disjoint in their effective support.*

Then there is a constant K_0 (depending only on the number of modules) and an aggregate locality parameter

$$\varepsilon_{IC} := K_0(\varepsilon_{\text{leak}} + \varepsilon_{\text{overlap}})$$

such that:

1. Small commutators. For any two contexts c, c' and any state Θ ,

$$d(U_c(U_{c'}(\Theta)), U_{c'}(U_c(\Theta))) \leq \varepsilon_{IC}.$$

2. Sequential vs. joint training. Let $c_1, \dots, c_T$ be a finite sequence of contexts and let

$$\Theta_{\text{seq}} := U_{c_T} \circ \dots \circ U_{c_1}(\Theta_0)$$

be the result of training on them once in that order. Let U_Σ denote an idealized joint update that optimally improves compression on the mixture $\frac{1}{T} \sum_t P_{c_t}$. Then

$$d(\Theta_{\text{seq}}, U_\Sigma(\Theta_0)) \leq O(T^2 \varepsilon_{IC}),$$

with Appendix A giving refined bounds in terms of the number of confusing pairs.

3. Forgetting bound. For any old context c that appears early in the sequence and any intermediate state Θ , the degradation in compression of P_c caused by a later update $U_{c'}$ satisfies

$$|L_c(U_{c'}(\Theta)) - L_c(\Theta)| \leq O(\varepsilon_{IC}),$$

and the total loss on P_c after training on $c_1, \dots, c_T$ is bounded by $O(T^2 \varepsilon_{IC})$. In particular, catastrophic forgetting is ruled out in the regime $\varepsilon_{IC} \rightarrow 0$ with moderate T .

Idea of proof. The context-locality and small-overlap assumptions imply that each U_c acts as an almost identity on modules outside S_c and as a local endomorphism on modules in S_c , with only $\varepsilon_{\text{leak}}$ disturbance elsewhere. This is an instance of the discrete CCL setting: the modules play the role of causal factors, the supports S_c are context supports, and $\varepsilon_{\text{leak}}$ and $\varepsilon_{\text{overlap}}$ bound the deviation from perfect factorwise updates.

The discrete CCL theorem then gives that the commutator $U_c \circ U_{c'}$ versus $U_{c'} \circ U_c$ is bounded by a constant times the leakage, yielding item (1). Standard estimates on compositions of non-expansive maps with small pairwise commutators give the $O(T^2 \varepsilon_{IC})$ bound between sequential and joint updates, and the forgetting bound follows by applying the mean value inequality to the context-wise losses L_c along the interpolating path between Θ_{seq} and $U_\Sigma(\Theta_0)$. The refined approximate categorical theorem shows how to replace the T^2 factor by a dependence on the number of confusing pairs.

Intuitively, updates for different contexts almost commute because they work on almost disjoint sets of modules, and where they do overlap, the clarity penalty (with meta-learned weights) ensures the incremental-compression overlap is small. This directly mirrors the neural-network story where small mixed Hessian blocks imply small gradient commutators. $\square$

8.6 Categorical viewpoint

Categorically, this example fits the general picture by taking:

- $\mathcal{P}$ to be the category whose objects are feature libraries, morphisms are incremental-compression update procedures (including their associated meta-learned clarity weights), and the monoidal product $\otimes$ is componentwise concatenation of modules;
- $\Theta \approx \Theta_1 \otimes \cdots \otimes \Theta_K$ to be the factorization of the feature library into modules;
- each U_c to be a context-local endomorphism that is (up to the small errors above) the identity on all factors $k \notin S_c$.

In the idealized zero-leakage limit $\varepsilon_{\text{leak}} = \varepsilon_{\text{overlap}} = 0$, each U_c literally factors as $\bigotimes_k U_c^{(k)}$ with $U_c^{(k)} = \text{id}$ for $k \notin S_c$, so the categorical GCL theorem applies and context updates with disjoint supports commute on the nose. The incremental-compression CCL theorem above then quantifies how this picture degrades gracefully when the supports are only approximately disjoint and overlaps are small but nonzero, and the meta-learned clarity weights act as an automatic regulator that keeps the system in the good regime.

8.7 One more step: McBride-inspired gradient guidance for causal incremental compression

We now outline one more possibly interesting step toward “causal coding”-izing incremental compression.

The incremental-compression update rule in the previous subsections treats the feature library Θ as a purely discrete object: at each context c the learner proposes a finite set of candidate edits (adding, removing, or modifying short features) and accepts those that improve a two-part code-length objective. In this subsection we sketch how to augment this search with a light-weight, “McBride-derivative-inspired” gradient and curvature estimate on library space, so that candidate edits are ranked not only by their raw local compression gain but by an approximate information-geometric score.

The formal justification for viewing feature libraries as living on a smooth manifold, and for defining a Fisher metric and natural gradient on this manifold using McBride-style derivatives of programs, is left to a companion paper. Here we only describe the practical pattern: how to use gradient-ish and Hessian-ish quantities to guide otherwise discrete incremental-compression moves.

Mathematical idea (informal). Fix a context c and a current library Θ . As before, let $L_c(\Theta)$ denote the expected two-part code length on data from context c when we run a fixed incremental-compression search using Θ .

We conceptually treat the library as a vector of coordinates

$$\Theta = (\theta_f)_{f \in \mathcal{F}},$$

with one coordinate θ_f per feature f (or per small group of features, or per module; the exact granularity is a design choice). For each feature f we estimate a “directional derivative”

$$g_c[f] \approx \frac{\partial L_c(\Theta)}{\partial \theta_f},$$

measuring how much L_c would change if we nudged the presence or strength of f . Concretely, we approximate $g_c[f]$ using compression gaps already computed by the incremental compressor, e.g. by comparing $L_c(\Theta)$ with the loss when f (or the module containing f) is disabled.

Similarly, we approximate a diagonal Fisher (or Gauss-Newton) term

$$G_c[f, f] \approx \mathbb{E}_{x \sim P_c} [(\partial_{\theta_f} \log p_{\Theta}(x))^2],$$

which heuristically measures curvature or “stiffness” of the loss in the direction of f . In practice we estimate $G_c[f, f]$ from the variability of the per-example compression effect of f , again using quantities the incremental compressor already computes.

Now consider a discrete candidate move m that edits the library in a small way: add a feature, remove a feature, or refine a feature inside some module. We associate to m a sparse “move vector” $\delta\Theta_m$ with components

$$\delta\Theta_m[f] = \begin{cases} +1 & \text{if } m \text{ adds or strengthens feature } f, \\ -1 & \text{if } m \text{ removes or weakens feature } f, \\ 0 & \text{otherwise.} \end{cases}$$

In these coordinates, the first-order change in L_c predicted by a Taylor expansion is

$$\Delta_c^{(1)}(m) \approx \nabla_{\Theta} L_c(\Theta)^\top \delta\Theta_m = \sum_f g_c[f] \delta\Theta_m[f].$$

The second-order term, using a diagonal Fisher / Gauss-Newton surrogate, is

$$\Delta_c^{(2)}(m) \approx \frac{1}{2} \delta\Theta_m^\top G_c(\Theta) \delta\Theta_m = \frac{1}{2} \sum_f G_c[f, f] (\delta\Theta_m[f])^2.$$

We then define a simple information-geometric score for move m as

$$\text{Score}_c(m) := -\Delta_c^{(1)}(m) - \lambda_{\text{curv}} \Delta_c^{(2)}(m) - \lambda_{\text{clar}} \text{ClarityPenalty}(m), \quad (1)$$

where $\lambda_{\text{curv}}, \lambda_{\text{clar}} \geq 0$ are hyperparameters and $\text{ClarityPenalty}(m)$ is any cross-context overlap penalty of the kind already used in the causal IC clarity term. Intuitively:

- $-\Delta_c^{(1)}(m)$ is the predicted first-order *benefit* of m for context c (measured in bits saved).
- $\Delta_c^{(2)}(m)$ penalizes moves that push strongly into directions where the loss is already highly curved or “stiff” in the Fisher sense.
- $\text{ClarityPenalty}(m)$ discourages edits that further entangle contexts or modules.

Ranking candidate discrete moves by $\text{Score}_c(m)$ and applying only those with positive score gives a purely combinatorial search that is nevertheless guided by a McBride-inspired natural gradient on the underlying program space.

Algorithmic recipe. We now sketch one possible implementation of the above idea, phrased entirely in terms of quantities an incremental compressor already maintains. The key point is that we never need to explicitly construct a continuous parameter manifold or a true McBride derivative; we only need per-feature compression statistics and simple proxy formulas.

We assume:

- A feature library $\Theta = (F^{(1)}, \dots, F^{(K)})$ factorized into modules, as before.

- A batch D_c of strings from context c .
- Existing code that, given Θ and D_c , runs an incremental compression search (e.g. greedy-with-refit) and returns, for each feature f used during compression on D_c , its net bits saved on that batch.

A single McBride-guided causal IC update for context c can then be written as follows.

Algorithm 1 Causal IC update with McBride-inspired gradient guidance

Require: Current library Θ , context c , data batch D_c

Require: Hyperparameters $\lambda_{\text{curv}}, \lambda_{\text{clar}} \geq 0$

Ensure: Updated library Θ'

1: Run the usual incremental compressor on D_c using Θ , obtaining:

- baseline code length estimate $L_c(\Theta)$;
- per-feature statistics: for each feature f , usage count and net bits saved on D_c .

2: For each feature f , estimate a gradient-like quantity

$$g_c[f] \approx -\text{BitsSaved}_c(f).$$

3: For each feature f , estimate a diagonal Fisher-like quantity $G_c[f, f]$, for example by

$$G_c[f, f] \approx \text{Average over strings in } D_c(g_c[f]^2).$$

4: Using the existing IC machinery, construct a finite set $\mathcal{M}_c$ of candidate discrete moves m (add, remove, or modify features and modules) that are locally promising according to standard incremental-compression heuristics.

5: **for** each move $m \in \mathcal{M}_c$ **do**

6: Define a sparse move vector $\delta\Theta_m[f]$ with entries $+1$ or -1 for features touched by m and 0 otherwise.

7: Compute the first-order gain

$$\text{lin_gain}(m) = - \sum_f g_c[f] \delta\Theta_m[f].$$

8: Compute the curvature cost (using diagonal Fisher)

$$\text{quad_cost}(m) = \frac{1}{2} \sum_f G_c[f, f] (\delta\Theta_m[f])^2.$$

9: Compute a clarity penalty $\text{ClarityPenalty}(m)$ by summing the existing cross-context overlap penalties over all features touched by m .

10: Define the McBride-inspired IG score

$$\text{Score}_c(m) = \text{lin_gain}(m) - \lambda_{\text{curv}} \text{quad_cost}(m) - \lambda_{\text{clar}} \text{ClarityPenalty}(m).$$

11: **end for**

12: Sort $\mathcal{M}_c$ in decreasing order of $\text{Score}_c(m)$.

13: **for** each move m in this sorted order **do**

14: **if** $\text{Score}_c(m) \leq 0$ **then**

15: **break**

▷ No further candidate move is predicted to help

16: **end if**

17: Tentatively apply m to Θ to obtain a candidate library Θ_m .

18: Optionally run a short local refit inside the affected modules of Θ_m on D_c .

19: Accept the move: set $\Theta \leftarrow \Theta_m$.

20: **end for**

21: **return** $\Theta' \leftarrow \Theta$

This procedure leaves the core incremental-compression machinery unchanged. It only adds a scoring layer on top of the usual candidate-move generator, using quantities that are already available (or cheap to compute) in a WILLIAM-style compressor. In this sense it is a “McBride-inspired gradient descent”: we treat discrete edits of a library of short programs as approximate steps along a hidden smooth manifold, and use a Fisher-style metric to decide which steps are most promising. The formal analytic details required to mathematically justify this sort of approach are more involved and are deferred to a separate information-geometric treatment of program spaces, but the algorithmic picture is relatively simple and we hope we have presented it clearly.

A A categorical formulation of causal continual learning

In this subsection we sketch a categorical formulation of the General Causal-Continual-Learning Theorem. The goal is to isolate the structural ingredients that make continual learning / continual adaptation possible, in a way that is agnostic to whether the underlying state spaces are continuous (manifolds, vector spaces) or discrete (finite sets, graphs). Both the continuous and discrete instances in the main text then appear as special cases of the same abstract pattern, and the approximate variants with explicit error bounds live naturally in this categorical setting.

A.1 Three layers of description as categories

We distinguish three levels of description, each naturally organized as a category:

- An *environment* category $\mathcal{E}$ whose objects are data-generating *contexts* E (e.g. different domains, datasets, prompts, goals, or interaction regimes), and whose morphisms encode coarse transformations between contexts (restriction, domain shift, conditioning, etc.).
- A category $\mathcal{C}$ of *causal models* (e.g. structural causal models or predictive-coding style generative models). Objects $C \in \mathcal{C}$ are causal descriptions of contexts (variables plus mechanisms); morphisms are structure-preserving maps between such models.
- A symmetric monoidal category $(\mathcal{P}, \otimes, I)$ of *parameterized learners*. Objects are parameter spaces (or state spaces) Θ , and morphisms $U : \Theta \rightarrow \Theta'$ are update procedures or learning transforms. The monoidal product $\otimes$ represents factorized composition of subsystems; I is the trivial (unit) system. Endomorphisms $\text{End}_{\mathcal{P}}(\Theta)$ model learning dynamics that map a parameter object back to itself.

The flow “environment $\rightarrow$ causal model $\rightarrow$ learner” is captured by functors

$$J : \mathcal{E} \rightarrow \mathcal{C}, \quad M : \mathcal{C} \rightarrow \mathcal{P}.$$

Given a context $E \in \mathcal{E}$, $J(E)$ is its causal model, and $M(J(E))$ is the corresponding learner-side object (e.g. a PC/CC network configured to operate in that context). For simplicity we will assume a fixed parameter object $\Theta \in \text{Ob}(\mathcal{P})$ shared across contexts, equipped with context-conditioned update endomorphisms $U_t : \Theta \rightarrow \Theta$.

A.2 Contexts as endomorphisms and continual learning as path-independence

A *context* t (in some index set of contexts) is represented at the learner level by an endomorphism

$$U_t : \Theta \rightarrow \Theta,$$

intuitively “update the learner once under context t with a small step size”. A continual-learning / continual-adaptation schedule for contexts $t_1, \dots, t_T$ then corresponds to the composite

$$U_{t_T} \circ \dots \circ U_{t_1} : \Theta \rightarrow \Theta.$$

In contrast, an idealized *joint multi-context* learner for a finite multiset of contexts $\{t_1, \dots, t_T\}$ would be represented by a single “all-at-once” update

$$U_\Sigma : \Theta \rightarrow \Theta,$$

obtained (informally) by training on, or adapting under, all contexts simultaneously. Catastrophic forgetting or interference is precisely the failure of *path-independence*: different orders of composing the U_t lead to substantially different effective maps, and these differ from U_Σ .

Categorically, good continual learning corresponds to the existence of canonical isomorphisms

$$\alpha_\sigma : U_{t_{\sigma(T)}} \circ \dots \circ U_{t_{\sigma(1)}} \Rightarrow U_\Sigma$$

for each permutation σ of $\{1, \dots, T\}$, satisfying the usual coherence conditions. In the strict case $\alpha_\sigma = \text{id}$ for all σ , the U_t form a commuting family inside the monoid $\text{End}_{\mathcal{P}}(\Theta)$:

$$U_{t_i} \circ U_{t_j} = U_{t_j} \circ U_{t_i} \quad \text{for all } i, j.$$

The analytic commutator-based results in the gradient-flow setting (Section 3) give a quantitative version of this statement in the smooth-manifold case, where the defect of commutativity is measured by Lie brackets of vector fields.

A.3 Causal factorization as monoidal structure

The categorical step where causality enters is the assumption that contexts admit a factorization into *causal components*, and that this factorization is respected functorially all the way down to the learner.

Formally, we assume:

- The causal-model category $\mathcal{C}$ is symmetric monoidal with product $\otimes$ and unit $I_{\mathcal{C}}$.
- For each context E , its causal model decomposes as

$$J(E) \cong C_1(E) \otimes \dots \otimes C_K(E),$$

where the $C_k(E)$ are “causal factors” (subsystems whose internal dynamics are richly coupled, but whose cross-factor couplings are weak or absent after appropriate interventions).

- The learner functor M is *strong monoidal*: there are natural isomorphisms

$$M(C \otimes C') \cong M(C) \otimes M(C'), \quad M(I_{\mathcal{C}}) \cong I$$

that are coherent in the usual sense. Thus

$$\Theta(E) := M(J(E)) \cong M(C_1(E)) \otimes \dots \otimes M(C_K(E)).$$

In words: causal factorization of a context induces a tensor factorization of the learner’s internal representation.

We will henceforth suppress the explicit dependence on E when it is clear from context, writing simply

$$\Theta \cong \Theta_1 \otimes \dots \otimes \Theta_K \quad \text{with} \quad \Theta_k := M(C_k).$$

A.4 Context-locality as factorwise endomorphisms

Causal coding is designed so that each context t “lives” on a subset of the causal factors. Categorically, this is expressed by:

Definition 3 (Context-local update). *Given a factorization $\Theta \cong \Theta_1 \otimes \cdots \otimes \Theta_K$, a context t is local to a subset $S_t \subseteq \{1, \dots, K\}$ if its update endomorphism $U_t : \Theta \rightarrow \Theta$ factors (up to the canonical associativity/symmetry isomorphisms of $\mathcal{P}$) as*

$$U_t \cong U_t^{(1)} \otimes \cdots \otimes U_t^{(K)},$$

where $U_t^{(k)} : \Theta_k \rightarrow \Theta_k$ and $U_t^{(k)} = \text{id}_{\Theta_k}$ for all $k \notin S_t$.

Intuitively, $U_t^{(k)}$ is the part of the update that adjusts the parameters devoted to causal factor k , and the identity elsewhere means that factor is left untouched when adapting under context t . Perfect causal disentangling would make S_t exactly the set of factors that are genuinely relevant for that context.

The central categorical observation is that the tensor product in a symmetric monoidal category automatically makes such context-local updates commute when their supports are disjoint.

Lemma 2 (Disjoint-support commutativity). *Let $(\mathcal{P}, \otimes, I)$ be a symmetric monoidal category, and $\Theta \cong \Theta_1 \otimes \cdots \otimes \Theta_K$. Suppose U_s and U_t are context-local endomorphisms with supports $S_s, S_t \subseteq \{1, \dots, K\}$ and $S_s \cap S_t = \emptyset$. Then*

$$U_s \circ U_t \cong U_t \circ U_s$$

via the canonical symmetry/associativity isomorphisms of $\mathcal{P}$. In particular, they commute strictly if we regard Θ as a strictly associative tensor and choose a fixed parenthesization.

Sketch. Up to canonical isomorphisms, we can write

$$U_s \cong \bigotimes_{k=1}^K U_s^{(k)}, \quad U_t \cong \bigotimes_{k=1}^K U_t^{(k)},$$

with $U_s^{(k)} = \text{id}$ for $k \notin S_s$ and $U_t^{(k)} = \text{id}$ for $k \notin S_t$. Because S_s and S_t are disjoint, for each factor k at least one of $U_s^{(k)}, U_t^{(k)}$ is the identity. Composition distributes factorwise:

$$U_s \circ U_t \cong \bigotimes_{k=1}^K (U_s^{(k)} \circ U_t^{(k)}), \quad U_t \circ U_s \cong \bigotimes_{k=1}^K (U_t^{(k)} \circ U_s^{(k)}).$$

If one of $U_s^{(k)}, U_t^{(k)}$ is id, the two compositions coincide on that factor; on all other factors both are the identity. By the symmetry and associativity coherence laws, these factorwise equalities assemble into an isomorphism $U_s \circ U_t \cong U_t \circ U_s$. In a strict monoidal presentation (Mac Lane strictification), this is an actual equality. $\square$

Thus, in the idealized case where causal coding succeeds perfectly and each context only adjusts the factors it truly depends on, we obtain a commuting family of context updates for all contexts that do not share causal support. The next subsection explains how to relax these conditions and quantify approximate commutativity in terms of small locality errors and sparse overlaps between supports.

A.5 Categorical General Causal-Continual-Learning Theorem

We can now state a purely categorical version of the general theorem.

Theorem 4 (Categorical GCL, factorized form). *Let $(\mathcal{P}, \otimes, I)$ be a symmetric monoidal category. Let $\Theta \in \text{Ob}(\mathcal{P})$ admit a factorization $\Theta \cong \Theta_1 \otimes \cdots \otimes \Theta_K$. Let $\{U_t : \Theta \rightarrow \Theta\}_{t \in T}$ be a family of context-conditioned update endomorphisms such that for each t :*

1. U_t is context-local with support $S_t \subseteq \{1, \dots, K\}$ in the sense above; and
2. the supports are causally minimal, i.e. no U_t contains within S_t a factor on which it acts as the identity while still being required to preserve performance in that context.

Then for any finite multiset $\{t_1, \dots, t_T\}$ whose supports are pairwise disjoint ($S_{t_i} \cap S_{t_j} = \emptyset$ for $i \neq j$), the corresponding updates commute (up to the canonical symmetry/associativity isomorphisms):

$$U_{t_{\sigma(T)}} \circ \cdots \circ U_{t_{\sigma(1)}} \cong U_{t_T} \circ \cdots \circ U_{t_1}$$

for every permutation σ of $\{1, \dots, T\}$. Consequently, there is a well-defined joint multi-context update

$$U_\Sigma := U_{t_T} \circ \cdots \circ U_{t_1}$$

independent (up to isomorphism) of the order in which the contexts are seen, i.e. no catastrophic forgetting or interference among these contexts.

Idea of proof. By Lemma 2, any pair U_{t_i}, U_{t_j} with disjoint supports commutes up to canonical isomorphism. The coherence axioms for symmetric monoidal categories ensure that these pairwise commutations extend to arbitrary finite products: any two permutations of a tensor of commuting endomorphisms are connected by a unique canonical isomorphism built out of symmetries and associators. Applying this to the family $\{U_{t_1}, \dots, U_{t_T}\}$ yields the claimed path-independence of their composite.

The causal minimality assumption rules out trivial factorizations where a context’s support is artificially enlarged by inserting identity actions on extra factors; this matters when we connect the abstract theorem back to the tradeoff results about independence versus explicit couplings and to the “weakness” principle. $\square$

In this categorical form, the role of causal modeling and causal coding is to *engineer* a factorization $\Theta \cong \bigotimes_k \Theta_k$ and context-local updates U_t that satisfy the hypotheses of the theorem as well as possible. The continuous-variable and discrete-variable specializations in the main text simply instantiate $\mathcal{P}$ as, respectively, a category of smooth manifolds with smooth maps and a category of finite sets (or graphs) with structure-preserving maps; the statement and the proof pattern remain the same at a structural level. In the next subsection we move to an approximate formulation, in which locality and commutativity are only satisfied up to explicit error terms and the conclusions degrade smoothly with these errors.

A.6 Connection to the tradeoff between preprocessing and disentangling

The categorical picture also makes precise how the Route A vs. Route B tradeoff (preprocessing for independence vs. internal disentangling) fits into continual learning across contexts.

- **Route A (preprocessing for independence).** Here we choose the environment-to-causal-model functor $J : \mathcal{E} \rightarrow \mathcal{C}$ so that $J(E)$ is as close as possible to a tensor product of weakly coupled factors, i.e. the causal graph for typical contexts is sparse and low-treewidth after appropriate interventions. Categorically, this is a choice of strong-monoidal structure on J that pushes correlation structure “upstream” into a small number of explicitly modeled couplings.
- **Route B (disentangling inside the learner).** Here we keep J relatively generic and instead invest in a strong monoidal functor $M : \mathcal{C} \rightarrow \mathcal{P}$ and in the architecture of $\mathcal{P}$ so that Θ acquires an internal factorization aligned with the causal factors. This corresponds to making the update endomorphisms U_t as block-diagonal as possible in the monoidal decomposition.

The tradeoff theorem says that any architecture attaining a given information-efficiency target must, in effect, pay for independence either via Route A (more aggressive transforms and factorization at the causal-model level) or Route B (more explicit multi-ary couplings and richer internal disentangling). In categorical terms, both routes are choices about how strongly monoidal the composite $M \circ J : \mathcal{E} \rightarrow \mathcal{P}$ can be made, relative to the underlying dependency structure of the environment.

From the standpoint of Theorem 4, the key quantity is how close the actual U_t are to perfectly context-local, factorwise endomorphisms. Route A improves this by simplifying the causal graph before it reaches the learner; Route B improves it by restructuring the learner so that its parameter object Θ realizes the causal product structure more faithfully. Causal coding is precisely the design discipline of using both routes, guided by interventional influence, to enlarge the class of contexts for which the hypotheses of the categorical GCL theorem are approximately satisfied, and hence to enlarge the regime where continual learning works without catastrophic forgetting.

A.7 Approximate categorical CCL: enriched setting and confusing pairs

In practice the hypotheses of the categorical Causal-Continual-Learning Theorem never hold exactly. Contexts are only approximately local to a subset of causal factors; supports only approximately disjoint; and the induced update maps commute only approximately. What we want is a statement of the form

If the factorization / locality / commutativity assumptions are satisfied up to small errors, then path-dependence and forgetting are also small, with a dependence on those errors that is controlled and not catastrophically sensitive.

This subsection sketches such an “approximate” version in a metric-enriched categorical setting, and shows how the global error scales (i) linearly in the local errors of the premises and (ii) linearly in the number of genuinely “confusing” context pairs. We then indicate how this plugs into the causal-coding neural-net and incremental-compression instantiations.

A.7.1 Metric-enriched monoidal setting and approximate locality

We work in a symmetric monoidal category $(\mathcal{P}, \otimes, I)$ that is enriched over Lawvere metric spaces. Concretely, we assume:

- For each pair of objects X, Y there is a (pseudo-)metric

$$d_{X,Y} : \text{Hom}_{\mathcal{P}}(X, Y) \times \text{Hom}_{\mathcal{P}}(X, Y) \rightarrow [0, \infty)$$

and we write d_X for $d_{X,X}$ on endomorphisms of X .

- Composition is non-expansive in each argument:

$$d_Z(V \circ U, V \circ U') \leq d_Y(U, U'), \quad d_Z(U \circ V, U' \circ V) \leq d_X(U, U')$$

whenever the composites are defined.

- The monoidal product is also non-expansive:

$$d_{X \otimes X', Y \otimes Y'}(U \otimes U', V \otimes V') \leq d_{X,Y}(U, V) + d_{X',Y'}(U', V').$$

As before, we fix a parameter object

$$\Theta \cong \Theta_1 \otimes \cdots \otimes \Theta_K$$

and regard $\text{End}_{\mathcal{P}}(\Theta)$ as a metric monoid with metric d_{Θ} . Contexts $c \in \mathcal{C}$ act at the learner level via endomorphisms

$$U_c : \Theta \rightarrow \Theta.$$

In the exact theorem, each U_c factors as a tensor product of per-factor maps that are identities outside a support $S_c \subseteq \{1, \dots, K\}$, and updates for contexts with disjoint supports commute strictly. To move to the approximate case, we introduce:

Definition 4 (Approximate context-locality). *Given a factorization $\Theta \cong \Theta_1 \otimes \cdots \otimes \Theta_K$, a context c has intended support $S_c \subseteq \{1, \dots, K\}$ if there exists an “ideal” local update*

$$V_c \cong V_c^{(1)} \otimes \cdots \otimes V_c^{(K)} : \Theta \rightarrow \Theta$$

with $V_c^{(k)} = \text{id}_{\Theta_k}$ for all $k \notin S_c$, such that the actual learner update U_c satisfies

$$d_{\Theta}(U_c, V_c) \leq \varepsilon_c.$$

We call $\varepsilon_c \geq 0$ the locality error for context c .

We also assume each U_c is non-expansive with respect to d_{Θ} (this is automatic if it is built from non-expansive primitives in $\mathcal{P}$).

A.7.2 Pairwise commutator defects and confusing pairs

The “defect of commutativity” for a pair of contexts c, c' at the ideal level is

$$\Delta_{c,c'} := d_{\Theta}(V_c \circ V_{c'}, V_{c'} \circ V_c).$$

In the exact factorized theorem, $\Delta_{c,c'} = 0$ whenever the supports S_c and $S_{c'}$ are disjoint. In practice, these defects are small but nonzero whenever contexts overlap weakly or the factorization is only approximate.

A first observation is that the commutator of the *actual* updates is controlled by the locality errors and the ideal commutator:

Lemma 3 (Approximate pairwise commutativity). *For any two contexts c, c' ,*

$$d_{\Theta}(U_c \circ U_{c'}, U_{c'} \circ U_c) \leq 2(\varepsilon_c + \varepsilon_{c'}) + \Delta_{c,c'}.$$

Sketch. Expand U_c and $U_{c'}$ around $V_c, V_{c'}$ and use non-expansiveness plus the triangle inequality:

$$\begin{aligned} d_{\Theta}(U_c \circ U_{c'}, U_{c'} \circ U_c) &\leq d_{\Theta}(U_c \circ U_{c'}, V_c \circ U_{c'}) + d_{\Theta}(V_c \circ U_{c'}, V_c \circ V_{c'}) \\ &\quad + d_{\Theta}(V_c \circ V_{c'}, V_{c'} \circ V_c) + d_{\Theta}(V_{c'} \circ V_c, V_{c'} \circ U_c) \\ &\quad + d_{\Theta}(V_{c'} \circ U_c, U_{c'} \circ U_c) \\ &\leq \varepsilon_c + \varepsilon_{c'} + \Delta_{c,c'} + \varepsilon_c + \varepsilon_{c'}, \end{aligned}$$

using non-expansiveness to pull $d_{\Theta}(U_{c'}, V_{c'})$ and $d_{\Theta}(U_c, V_c)$ out of composites. Collecting terms gives the claim. $\square$

This suggests defining a per-pair *confusion cost*

$$\delta_{c,c'} := 2(\varepsilon_c + \varepsilon_{c'}) + \Delta_{c,c'},$$

and calling a pair (c, c') *confusing* if $\delta_{c,c'}$ exceeds some small threshold τ_{pair} . Intuitively, confusing pairs are ones where either the contexts share significant causal support or the approximate factorization is poor.

A.7.3 Approximate CCL with linear dependence on confusing pairs

Consider a finite multiset of contexts

$$\Sigma = \{c_1, \dots, c_T\}$$

and the associated family of updates $U_{c_i} : \Theta \rightarrow \Theta$. For any permutation σ of $\{1, \dots, T\}$, write

$$U_{\sigma} := U_{c_{\sigma(T)}} \circ \dots \circ U_{c_{\sigma(1)}}.$$

Fix once and for all a *canonical* ordering of contexts (for instance by some index) and let σ_{can} be the permutation that lists the multiset Σ in that canonical order; write $U_{\text{can}} := U_{\sigma_{\text{can}}}$.

The key combinatorial fact is that any permutation can be transformed into the canonical one by a sequence of adjacent swaps, and we can choose this sequence so that every unordered pair of indices (i, j) is swapped at most once (this is just the usual sorting-by-inversions picture). Using Lemma 3, each adjacent swap of U_{c_i} and U_{c_j} changes the overall composite by at most δ_{c_i, c_j} in the metric d_{Θ} . Summing over the swaps then gives:

Theorem 5 (Approximate categorical CCL with confusing pairs). *Let $(\mathcal{P}, \otimes, I)$ be a symmetric monoidal category enriched over Lawvere metric spaces as above, and let $\Theta \cong \Theta_1 \otimes \dots \otimes \Theta_K$. Suppose each context c has an intended support S_c and an ideal local update V_c with locality error ε_c , and define confusion costs $\delta_{c,c'}$ as above.*

For any finite multiset of contexts $\Sigma = \{c_1, \dots, c_T\}$, let

$$E_{\text{conf}}(\Sigma) := \{\{i, j\} : 1 \leq i < j \leq T, \delta_{c_i, c_j} > 0\}$$

be the set of unordered confusing pairs among these contexts. Then there is a constant $C_{\mathcal{P}}$ (depending only on the enrichment constants of $\mathcal{P}$) such that for any permutation σ ,

$$d_{\Theta}(U_{\sigma}, U_{\text{can}}) \leq C_{\mathcal{P}} \sum_{\{i, j\} \in E_{\text{conf}}(\Sigma)} \delta_{c_i, c_j}.$$

In particular, if we put a uniform bound $\delta_{c_i, c_j} \leq \bar{\delta}$ and write $M := |E_{\text{conf}}(\Sigma)|$ for the number of confusing pairs, then

$$d_{\Theta}(U_{\sigma}, U_{\text{can}}) \leq C_{\mathcal{P}} \bar{\delta} M,$$

so the path-dependence error grows at most linearly in the number of confusing pairs and linearly in the pairwise commutator/locality errors.

Idea of proof. As in the exact case, the composite U_σ changes only when adjacent factors are swapped. For any adjacent swap exchanging U_{c_i} and U_{c_j} , Lemma 3 and non-expansiveness of pre- and post-composition give

$$d_\Theta(\cdots \circ U_{c_j} \circ U_{c_i} \circ \cdots, \cdots \circ U_{c_i} \circ U_{c_j} \circ \cdots) \leq C_{\mathcal{P}} \delta_{c_i, c_j}$$

for some fixed $C_{\mathcal{P}}$ that depends only on the enrichment constants. Now sort σ to σ_{can} by adjacent swaps using any sorting algorithm that realizes each inversion pair (i, j) at most once (for example: repeatedly bubble the correct element into position 1, then 2, etc.; once an element is in place it never moves again).

Each unordered pair $\{i, j\}$ that is already in canonical relative order is never swapped; each pair that is out of order is swapped exactly once. The total change in the composite between σ and σ_{can} is thus bounded by the sum of $C_{\mathcal{P}} \delta_{c_i, c_j}$ over those pairs that are actually swapped, which is a subset of $E_{\text{conf}}(\Sigma)$. This gives the stated bound. $\square$

Conceptually: the approximate categorical premises amount to small locality errors ε_c and small ideal commutator defects $\Delta_{c, c'}$ for most pairs. The theorem says that the resulting path-dependence and forgetting error depends smoothly (indeed, linearly) on those premise-errors, and only for the pairs of contexts that actually interact in a confusing way. In the worst case, when every pair is confusing, $M = O(T^2)$ and we recover the familiar quadratic dependence. But in many realistic regimes the confusion graph is sparse, and M scales like $O(T)$ or $O(T \log T)$ instead.

Probabilistic variants are immediate: if the ε_c and $\Delta_{c, c'}$ are random with small expectations (or tails), then linearity plus the triangle inequality gives corresponding expectation and high-probability bounds on $d_\Theta(U_\sigma, U_{\text{can}})$.

A.7.4 Connection to causal-coding neural networks

In the PC/CC neural-network instantiation, we can take:

- Objects of $\mathcal{P}$ to be parameter manifolds (or Euclidean parameter spaces) equipped with a metric d_Θ that measures performance distance, for example a supremum over per-task loss differences:

$$d_\Theta(\theta, \theta') := \sup_{c \in \mathcal{C}} |\mathcal{L}_c(\theta) - \mathcal{L}_c(\theta')|.$$

- Morphisms to be update operators (gradient steps, PC-style inference/learning updates), and enrichment to come from Lipschitz continuity of the training dynamics under this metric.
- The factorization $\Theta \cong \Theta_1 \otimes \cdots \otimes \Theta_K$ to be a block decomposition into modules (layers, heads, slots, routing blocks) engineered by causal coding.
- Locality errors ε_c to measure how much updating on context c perturbs layers that have negligible do-influence for c (for example, via norms of gradients or influence-function estimates on non-support modules).
- Ideal commutator defects $\Delta_{c, c'}$ (or their surrogates) to be controlled by mixed Hessian blocks or Lie brackets of gradient vector fields:

$$[g_c, g_{c'}] \approx 0 \quad \Rightarrow \quad d_\Theta(U_c \circ U_{c'}, U_{c'} \circ U_c) \text{ small.}$$

From this viewpoint, the approximate categorical theorem says: if causal coding succeeds in making cross-context mixed Hessians small outside a sparse pattern of genuinely interacting context pairs, and if updates on a context mostly stay within its do-influence support, then sequential PC/CC training behaves almost like joint training. The deviation scales linearly with (i) the magnitude of the mixed Hessian / commutator terms and (ii) the number of context pairs for which those terms are non-negligible.

Operationally, this gives an architecture- and data-dependent way to estimate how far one is from the ideal continual-learning regime: measure empirical ε_c and pairwise commutator proxies for a sample of contexts, build an approximate confusion graph, and then plug those into Theorem 5 to bound the expected path-dependence of training.

A.7.5 Connection to incremental compression and IC-based causal coding

In the incremental-compression setting, we already have an explicit metric

$$d(\Theta, \Theta') := \sup_{c \in \mathcal{C}} |L_c(\Theta) - L_c(\Theta')|$$

on feature libraries, and the context updates U_c were assumed 1-Lipschitz with respect to d . The factorization $\Theta \cong \Theta_1 \otimes \cdots \otimes \Theta_K$ is given by the modular decomposition of the feature library, and the context-locality and small-overlap assumptions translate directly into:

- locality errors ε_c controlled by leakage outside the support modules ($\varepsilon_{\text{leak}}$ in the IC-CCL theorem);
- pairwise commutator bounds $\delta_{c,c'}$ controlled by cross-context incremental compression overlap ($\varepsilon_{\text{overlap}}$ and the clarity penalty).

The “Incremental-compression CCL” theorem can thus be read as a concrete instance of Theorem 5: the metric-enriched category is the category of feature libraries with d -non-expansive incremental-compression updates; the supports S_c are given by do-influence on modules; and the confusion graph encodes which pairs of contexts actually share significant compressive structure. The linear-in- T^2 bounds in the earlier discrete statement are the worst-case specialization where every pair is potentially confusing. When the confusion graph is sparse, applying the approximate categorical theorem tightens this to linear dependence on the number of confusing pairs, exactly as in the neural setting.

In both the neural and IC cases, the moral is the same: causal coding tries to engineer a world in which (i) most contexts are approximately local to small supports, and (ii) only a sparse set of context pairs share substantial causal/compressive structure. The approximate categorical CCL theorem then guarantees that, in such a world, continual learning is robust: the errors induced by approximate factorization and nonzero commutators accumulate in a controlled, essentially linear way, rather than exploding combinatorially with the total number of contexts.

B Illustrative example: causal coding in a PC Transformer with overlapping textual contexts

Here we walk through a hypothetical example of causal coding in a PC transformer, at a moderate level of detail.

In real life, a single prompt can blend instructions, narrative, mathematics, programming, and social chit-chat. So one cannot think about discrete task types but needs to think in terms of

contexts: soft, overlapping regions in a high-dimensional space of situations that shape what counts as good prediction.

Here we sketch how causal coding might operate in a PC Transformer in the presence of such overlapping contexts, and how this still fits the general causal-continual-learning picture.

B.1 Two fuzzy, overlapping contexts

Consider a PC Transformer trained on mixed internet text. Among many possible contexts, focus on two qualitative ones:

- a *technical-math* context c_{math} in which the model is expected to manipulate formulas, perform small derivations, and keep symbols consistent;
- a *conversational-explanation* context c_{explain} in which the model is expected to speak in a friendly, informal tone and adapt to the user’s wording.

These are not disjoint: a prompt like

“Can you walk me through how to solve a quadratic equation step by step, but in a casual way as if you were explaining to a friend?”

sits in the overlap. It simultaneously demands mathematically correct symbolic reasoning (pulling toward c_{math}) and informal, dialogue-like phrasing (pulling toward c_{explain}).

We can formalize this by associating to each input sequence x a *context embedding* $c(x) \in \mathbb{R}^d$ and a soft context mixture $w_c(x)$ over a small dictionary of prototypical contexts $\{c_\alpha\}_{\alpha \in \mathcal{A}}$:

$$w_\alpha(x) \geq 0, \quad \sum_{\alpha \in \mathcal{A}} w_\alpha(x) = 1, \quad c(x) \approx \sum_{\alpha} w_\alpha(x) c_\alpha.$$

For our running example, $\mathcal{A}$ would contain at least c_{math} and c_{explain} , and the prompt above would yield some nontrivial mixture $w_{\text{math}}(x), w_{\text{explain}}(x) \in (0, 1)$.

B.2 Context-conditioned do-influence at the head/channel level

In a PC Transformer, each layer ℓ has multiple self-attention heads $h = 1, \dots, H_\ell$ and feed-forward channels. Predictive coding endows these with latent “value” nodes $z^{(\ell, h)}$ and local error units $\varepsilon^{(\ell)}$, updated via inference dynamics until a fixed point is reached.

Causal coding adds an interventional layer to this picture:

1. For a given input x , run PC inference to an equilibrium $\{z^{(\ell, h)}(x), \varepsilon^{(\ell)}(x)\}$.
2. Define *context-conditioned error functionals* that emphasize different aspects of the prediction quality. For example,

$$L_{\text{math}}(x) = \text{loss on tokens classified as formulas / code / numeric},$$

$$L_{\text{explain}}(x) = \text{loss on tokens in natural-language explanations, style constraints, etc.}$$

These are not separate training runs; they are different ways of reading the same prediction trace through different lenses.

3. For each head (ℓ, h) and each prototypical context $\alpha \in \mathcal{A}$, estimate an interventional influence

$$C_{\ell,h}^{(\alpha)} \approx \left. \frac{\partial}{\partial \delta} \right|_{\delta=0} L_{\alpha} \left(\text{equilibrium under } \text{do}(z^{(\ell,h)} \leftarrow z^{(\ell,h)} + \delta) \right).$$

In practice, $C_{\ell,h}^{(\alpha)}$ is obtained via a Schur-Fisher, natural-gradient, or Schrödinger-bridge surrogate rather than literal finite differences, but conceptually it measures: “if I nudge this head, how much does context- α loss change after PC relaxation?”

4. Turn these influences into *context-specific gates* on head (ℓ, h) :

$$r_{\ell,h}^{(\alpha)} = \frac{|C_{\ell,h}^{(\alpha)}|^p}{\sum_{h'} |C_{\ell,h'}^{(\alpha)}|^p + \kappa}, \quad p \in [1, 2], \kappa > 0.$$

Now, when we process a particular input x with soft context weights $w_{\alpha}(x)$, we define an *effective gate* for each head:

$$r_{\ell,h}(x) = \sum_{\alpha \in \mathcal{A}} w_{\alpha}(x) r_{\ell,h}^{(\alpha)}.$$

These gates modulate both inference (how much the head’s latent activity is allowed to move in response to errors) and learning (how strongly we update that head’s parameters on this example).

Intuitively, a “math-logic” head that carries strong influence on L_{math} but little on L_{explain} will have $r_{\ell,h}^{(\text{math})} \gg r_{\ell,h}^{(\text{explain})}$. For a purely conversational prompt, $w_{\text{math}}(x) \approx 0$, so that head is mostly frozen. For a purely technical derivation, $w_{\text{math}}(x) \approx 1$, so the head learns aggressively. For a blended prompt, it receives an intermediate weight.

B.3 Overlapping contexts and approximate modularity

Crucially, we never assume that contexts are disjoint or that examples belong to a single context. Instead:

- Each context α exposes one *direction* in loss space, given by L_{α} , and one pattern of head/channel influence $C_{\ell,h}^{(\alpha)}$.
- Causal coding attempts to find a representation in which each context α is supported mostly by a relatively small subset of heads/channels (its “support”), and in which mixed influence terms across very different contexts are small (weak cross-Hessian blocks).
- The soft mixture $w_{\alpha}(x)$ simply interpolates between these context-specific supports at training time.

In a mature model we might see, for example:

- A cluster of heads in middle layers that are heavily used whenever $w_{\text{math}}(x)$ is large (they encode algebraic patterns, symbol matching, and numeric regularities), and are mostly gated off for conversational small talk.
- Another cluster of heads that activate whenever $w_{\text{explain}}(x)$ is large (they track pronouns, politeness, discourse markers, etc.), and are only weakly involved in raw formula manipulation.

- A small overlapping subset of “bridge” heads that respond to both contexts: they know how to wrap a derivation in a friendly explanation or how to embed a formula inside a sentence. Their influence appears in both $C^{(\text{math})}$ and $C^{(\text{explain})}$, but even here causal coding tries to keep their effects disentangled (e.g. separate subchannels for content vs style).

From the perspective of the General Causal-Continual-Learning Theorem, each prototypical context α defines an update operator U_α (or a gradient field X_α) corresponding to optimizing L_α with context-specific gates. The causal-coding machinery pushes these operators to be *context-local*: U_α mostly adjusts the head/channel modules that have strong $C^{(\alpha)}$ influence, leaving others nearly fixed. This is exactly the “small residual gradient and small cross-Hessian” condition in the manifold version of the theorem, or the “almost identity outside support” condition in the discrete and categorical versions, up to the approximate error terms made explicit in the enriched categorical setting.

In the overlapping example:

- On pure math prompts, U_{math} is active and mostly acts on the math-logic heads; U_{explain} is nearly the identity.
- On pure conversational prompts, the roles reverse.
- On blended prompts, both operators act, but because their supports are largely distinct (with a small, carefully-structured overlap), their gradient flows almost commute. Doing a “math update” then an “explanation update” is nearly the same as doing them in the opposite order, and both are close to a joint step on $L_{\text{math}} + L_{\text{explain}}$. The residual non-commutativity is controlled by the pairwise confusion costs between these two contexts.

Thus, even without any hard partition of training into separate tasks, the model can achieve continual-learning-like robustness with respect to *contexts*: improving its behavior for one slice of the prompt space (e.g. technical-math) will not dramatically degrade its behavior for another slice (e.g. conversational explanation), as long as causal coding keeps their effective modules weakly entangled and the confusion graph remains sparse.

B.4 Why this does not require discrete context labels

The story above used two named contexts for clarity, but the machinery does not require explicit labels. In practice one can:

- maintain a small bank of context prototypes c_α discovered unsupervised (for example via clustering of prompts or of error patterns);
- define L_α by reweighting the same underlying token-level loss with attention to different subsets of tokens or features;
- estimate do-influences $C^{(\alpha)}$ and gates $r^{(\alpha)}$ for these prototypes, then let $w_\alpha(x)$ be inferred by a learned context encoder.

The key requirement for the general theorem is not that the contexts be named or sharply defined, but that the resulting family of context-conditioned update operators $\{U_\alpha\}$ be approximately local in a shared causal factorization of the Transformer. Causal coding at the head and channel level is precisely the mechanism that tries to enforce this: it continuously re-arranges and prunes the internal structure so that context-conditioned gradients stay as factorized as the data allow, making the corresponding flows nearly commuting and, therefore, making context-wise continual learning robust in the approximate sense formalized by the enriched categorical CCL theorem.

C Asymptotic equivalence and acceleration of causal IC

We now explore the nature of the advantage obtained by augmenting standard incremental compression with causal coding type methodologies. Throughout this section we explicitly restrict attention to a *best-case* regime in which there exists a feature library that is both globally optimal for plain incremental compression and perfectly modular according to the clarity penalty. Outside this regime, plain IC and causal IC will in general converge to different optima, and the analysis here does not apply directly; instead, causal IC then trades off some raw compression performance for improved modularity and continual-learning behavior.

At a high level, incremental compression with causal coding (IC+CC) is meant to change *how* the learner gets to its long-run feature library, not *what* it can ultimately represent.

- A plain IC learner maintains a single global feature library and keeps adding/refitting features to minimize expected two-part code length across all contexts. Nothing stops it from letting the same feature do double duty for multiple unrelated contexts, or from letting older features slowly drift as it tries to accommodate new contexts.
- A causal IC learner uses do-influence and clarity penalties to nudge the search toward *modular* libraries: features are grouped into modules that mainly serve particular context families (plus possibly a “shared” module), and cross-context, cross-module overlap is actively discouraged.

If the environment really has an underlying modular structure, there should be (at least one) representation that is both *globally code-length optimal* and *perfectly modular*. In that case, adding a clarity penalty does not change what the best long-run solution is; it only makes the learner prefer paths that get there via modular libraries instead of entangled ones. The picture we want is:

Plain IC and causal IC converge (in the limit) to the *same* optimal incremental code, but causal IC converges *faster* in a multi-context setting because the clarity penalty improves the local curvature (condition number) of the optimization problem and shrinks cross-context interference.

Below we state a simple version of this, in the same spirit as the discrete/categorical CCL theorem, but now with an explicit comparison between plain IC and IC+CC. We emphasize again that this theorem is conditional on the existence of a modular optimum that is also globally optimal for plain IC; in more general situations, regularization by R moves the optimum and the leading effect of IC+CC is to bias the learner toward different, more modular solutions.

Setup: average loss and clarity penalty. Let $\mathcal{C}$ be a family of contexts c , each with a distribution P_c over $\{0, 1\}^N$. Let π be a mixing distribution over contexts (the long-run frequency with which contexts appear). A feature library state is $\Theta = (\Theta_1, \dots, \Theta_K)$ where each Θ_k is a module as in the previous subsection.

For a library Θ and context c , let $L_c(\Theta)$ be the expected two-part description length on P_c given Θ , using some fixed incremental-coding search (greedy-with-refit, beam+refit, or PGM-guided search). Define the long-run average code length

$$F(\Theta) := \mathbb{E}_{c \sim \pi}[L_c(\Theta)].$$

To measure lack of modularity across contexts, we use a clarity penalty $R(\Theta)$ of the kind introduced earlier: for each feature f we track per-context gains $\gamma_c(f)$ and set

$$\text{Overlap}(f) := \sum_{c \neq c'} \min\{\gamma_c(f), \gamma_{c'}(f)\}, \quad R(\Theta) := \sum_{f \in \Theta} \text{Overlap}(f),$$

optionally normalized. Intuitively, $R(\Theta)$ is large when many features carry significant mass for many different, potentially unrelated contexts.

We consider two learners on the same environment:

- **Plain IC.** Updates the library by approximately minimizing $F(\Theta)$ (via incremental search and refitting), with no explicit clarity penalty.
- **Causal IC (IC+CC).** Updates the library by approximately minimizing

$$F_\lambda(\Theta) := F(\Theta) + \lambda R(\Theta)$$

for some fixed $\lambda > 0$ (or a slowly varying schedule λ_t). Gates and freezing outside the context support $S_c(\Theta)$ implement the “approximately minimizing F_λ ” part, as in the previous incremental-CC update rule.

We compare the long-run behaviour of these two learners under the following idealized assumptions.

Assumptions.

1. **SF-HGM with modular optimum.** Strings in each context c are generated by a short-feature hierarchical generative model (SF-HGM) with cross-information bound δ , so that there exist high-efficiency incremental decompositions with short features and small overlaps. Moreover, there exists at least one library Θ^* such that:
 - Θ^* globally minimizes $F(\Theta)$;
 - Θ^* is perfectly modular in the sense that all strong features for context c live in its support modules $S_c(\Theta^*)$, and each feature either serves a small family of related contexts or a dedicated “shared” module;
 - this modularity is exactly what R rewards: $R(\Theta^*) = 0$ and $R(\Theta) > 0$ for all $\Theta \neq \Theta^*$ with $F(\Theta) \leq F(\Theta^*) + O(\log N)$.
2. **Smoothness and local strong convexity.** In a neighbourhood N of Θ^* , the average loss $F(\Theta)$ admits a smooth surrogate $\tilde{F}(\Theta)$ (think: differentiable meta-objective over feature parameters) with L -Lipschitz gradient and strong convexity constant $\mu > 0$. Likewise R admits a smooth surrogate $\tilde{R}$ with L_R -Lipschitz gradient and Hessian $\nabla^2 \tilde{R}(\Theta^*) \succeq \nu I$ on the “entanglement directions” (those that change cross-context overlaps but not F to first order).
3. **Shared optimum.** For all λ in a range $[0, \lambda_{\max}]$, the combined objective $\tilde{F}_\lambda := \tilde{F} + \lambda \tilde{R}$ has the *same* unique minimizer Θ^* on N . (This is enforced by the “modular optimum” assumption: Θ^* is the only point that simultaneously minimizes F and R in that region.)
4. **IC updates approximate gradient steps.** Both plain IC and IC+CC implement stochastic update rules of the form

$$\begin{aligned}\Theta_{t+1} &= \Theta_t - \eta_t(\nabla \tilde{F}(\Theta_t) + \xi_t), \\ \Theta_{t+1}^{\text{cc}} &= \Theta_t^{\text{cc}} - \eta_t(\nabla \tilde{F}(\Theta_t^{\text{cc}}) + \lambda \nabla \tilde{R}(\Theta_t^{\text{cc}}) + \xi_t^{\text{cc}}),\end{aligned}$$

where $\{\xi_t\}$ and $\{\xi_t^{\text{cc}}\}$ are martingale-noise terms coming from finite samples and approximate search, with bounded variance; and the step sizes satisfy standard stochastic-approximation conditions $\sum_t \eta_t = \infty$, $\sum_t \eta_t^2 < \infty$.

Theorem (Asymptotic equivalence and acceleration of causal IC). Under Assumptions 1-4 above, and for any fixed $\lambda \in (0, \lambda_{\max}]$, the following hold:

1. **Same asymptotic optimum.** Both plain IC and causal IC converge (in probability, or almost surely under standard stochastic-approximation conditions) to the same modular optimum Θ^* :

$$\Theta_t \rightarrow \Theta^*, \quad \Theta_t^{\text{cc}} \rightarrow \Theta^*.$$

In particular, in the long run IC+CC learns the same SF-HGM structure as plain IC (up to the usual incremental-compression non-uniqueness). This *does not* hold in general; it is specific to the special case where Θ^* simultaneously minimizes both F and R .

2. **Improved local curvature.** In the neighbourhood N , the Hessian of $\tilde{F}_\lambda$ satisfies

$$\nabla^2 \tilde{F}_\lambda(\Theta) = \nabla^2 \tilde{F}(\Theta) + \lambda \nabla^2 \tilde{R}(\Theta) \succeq (\mu + \lambda\nu)I$$

on the entanglement directions, while its gradient is $(L + \lambda L_R)$ -Lipschitz. Thus $\tilde{F}_\lambda$ has an improved condition number

$$\kappa_{\text{cc}} := \frac{L + \lambda L_R}{\mu + \lambda\nu} \leq \frac{L}{\mu} =: \kappa_{\text{plain}},$$

with strict inequality whenever $\nu > 0$ and $\lambda > 0$.

3. **Faster linear convergence rate (in the noise-free limit).** Consider for simplicity noiseless gradient descent with a fixed step size $\eta \in (0, 2/(L + \lambda L_R))$. Then

$$\begin{aligned} \|\Theta_{t+1} - \Theta^*\| &\leq (1 - \eta\mu) \|\Theta_t - \Theta^*\|, \\ \|\Theta_{t+1}^{\text{cc}} - \Theta^*\| &\leq (1 - \eta(\mu + \lambda\nu)) \|\Theta_t^{\text{cc}} - \Theta^*\|. \end{aligned}$$

Since $\mu + \lambda\nu > \mu$, the causal-IC iterates contract toward Θ^* strictly faster:

$$\|\Theta_t^{\text{cc}} - \Theta^*\| \lesssim \exp(-t\eta(\mu + \lambda\nu)), \quad \|\Theta_t - \Theta^*\| \lesssim \exp(-t\eta\mu).$$

With stochastic noise present, the same comparison holds up to the usual $O(\sqrt{\eta t})$ fluctuation terms.

Interpretation. The assumptions are idealized and reflect a best-case regime:

- SF-HGM plus “modular optimum” says that there really is a representation in which each context family uses its own subset of modules, with only genuinely shared structure kept in shared modules, and that this representation is already near-optimal in plain IC’s sense. This is not guaranteed in general; when no such perfectly modular optimum exists, plain IC and IC+CC will converge to (typically different) local optima that trade off raw compression against modularity.
- The clarity penalty R is chosen so that it is minimized exactly at such modular representations and penalizes cross-context, cross-module overlap. Thus, under Assumption 3, adding λR does not move the location of the best solution, it just increases curvature in directions that correspond to entangling contexts.

- In that regime, IC+CC behaves like a preconditioned version of plain IC: it has the same fixed point Θ^* , but the local condition number of its objective is smaller, so gradient-like updates converge faster in the multi-context setting.

Intuitively: if you run plain incremental compression long enough on many contexts, you will, under SF-HGM assumptions and the existence of a modular global optimum, eventually discover a good hierarchy of features. Adding causal coding and clarity *does not* change what the best hierarchy is under those assumptions; it just biases the search so that features which mix unrelated contexts are disfavored early, and context-local modules emerge faster. The CCL-style almost-commuting update results then give you better continual-learning behavior en route, on top of this accelerated convergence. Outside this special regime, IC+CC converges to a different, more modular optimum than plain IC, and the analysis here should be read as a local picture around hypothetical modular solutions rather than a universal convergence claim.